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(54) **MEDICATION CARTRIDGE**

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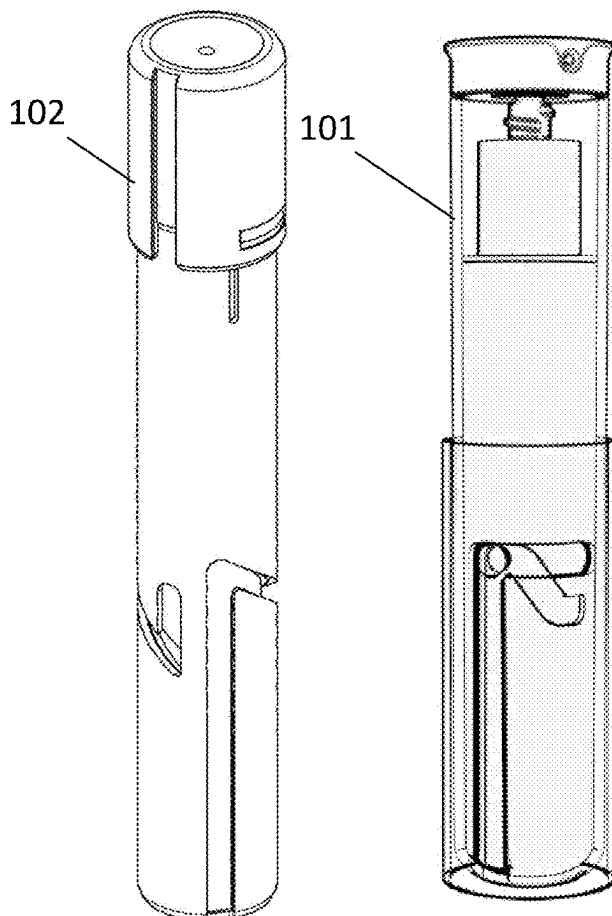
tion No. 18/606,136, filed on Mar. 15, 2024, now Pat. No. 12,106,836, Continuation-in-part of application No. 18/606,151, filed on Mar. 15, 2024, Continuation-in-part of application No. 18/606,173, filed on Mar. 15, 2024.

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(57) **ABSTRACT**

The present disclosure provides a medication cartridge and associated dispensing device. The dispensing device includes a housing containing a cartridge carrier, designed with a specific connector to align with a medication cartridge. The medication cartridge includes a shell with a medicine vial slidably positioned therein. When the dispensing device is activated, the cartridge carrier rotates to raise the vial and cause the vial to dispense medicine from the vial and out of a dispensing aperture for use by a patient.



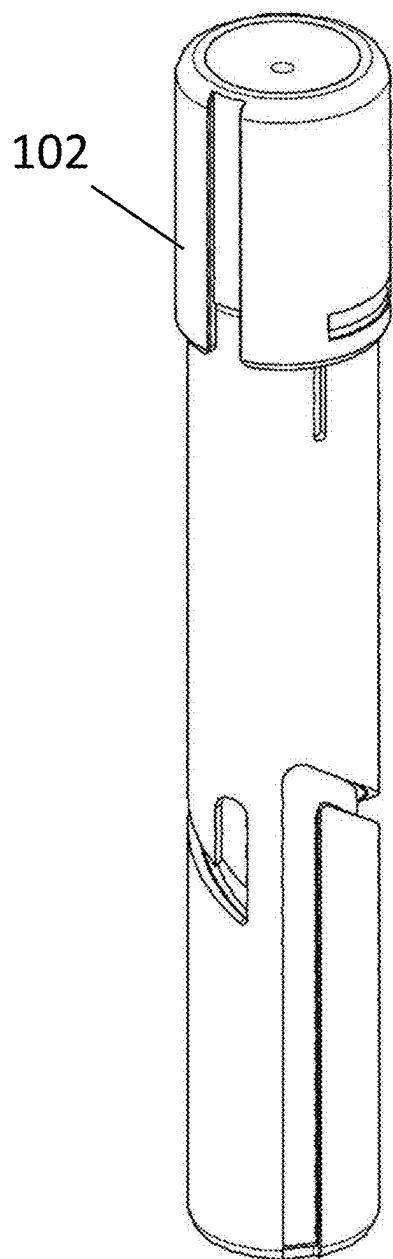


FIG. 1A

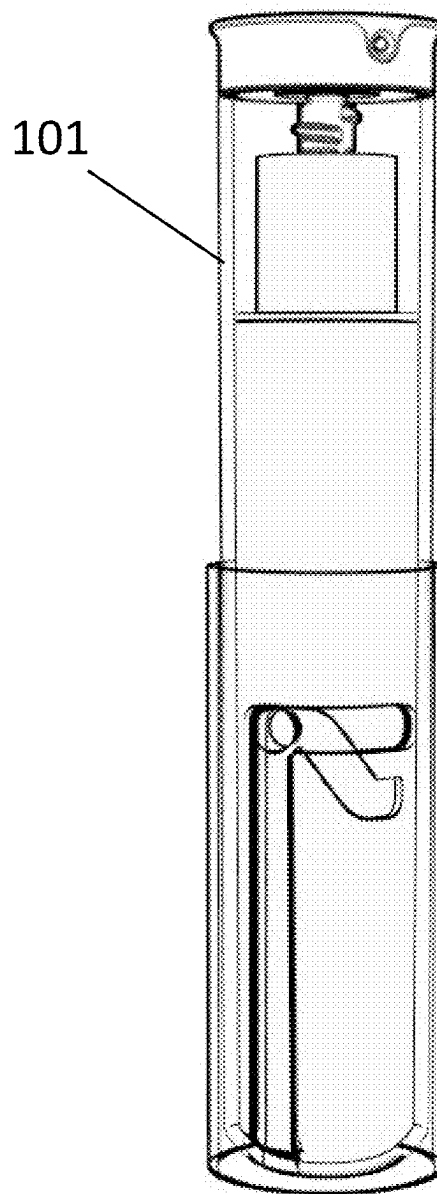


FIG. 1B

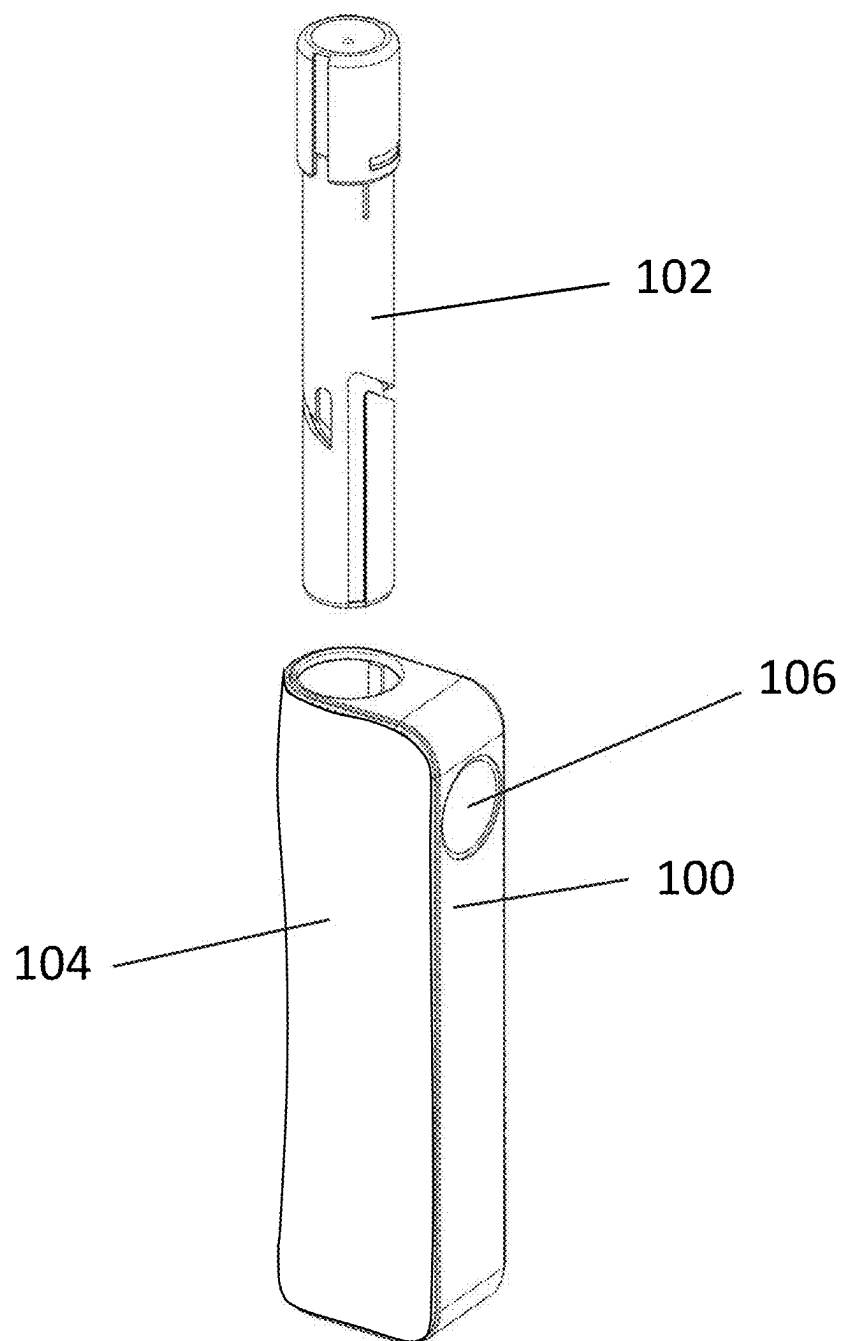


FIG. 1C

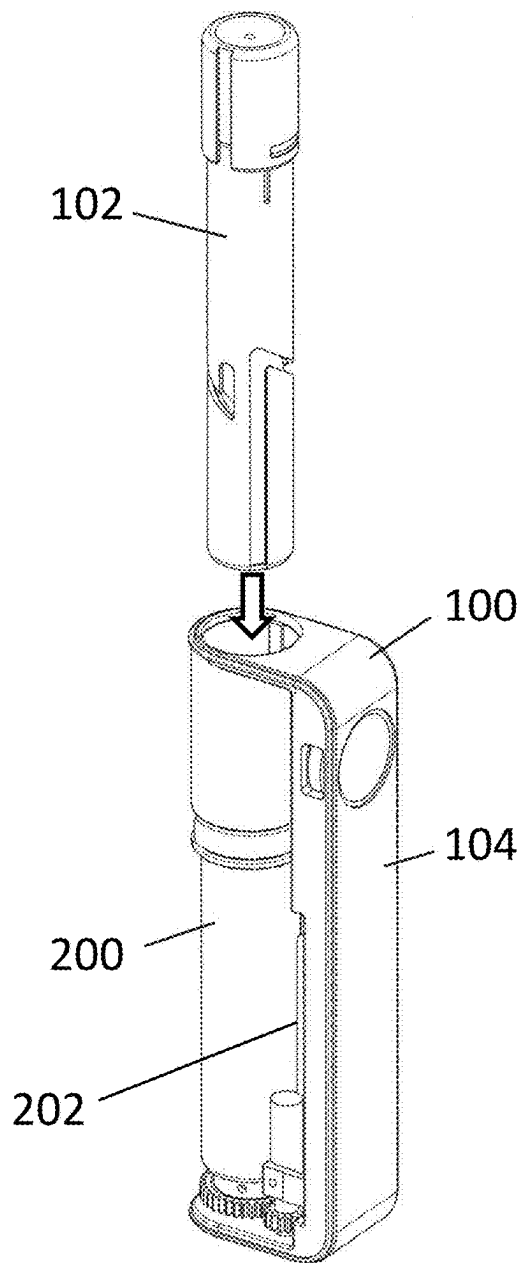


FIG. 2A

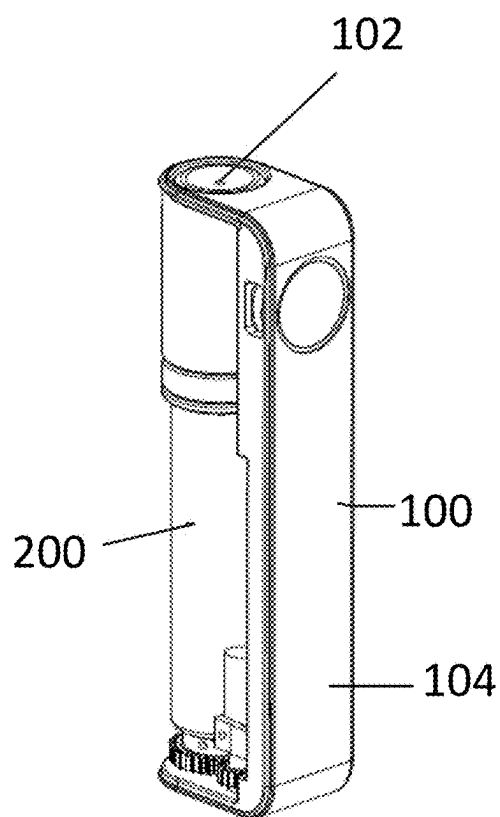


FIG. 2B

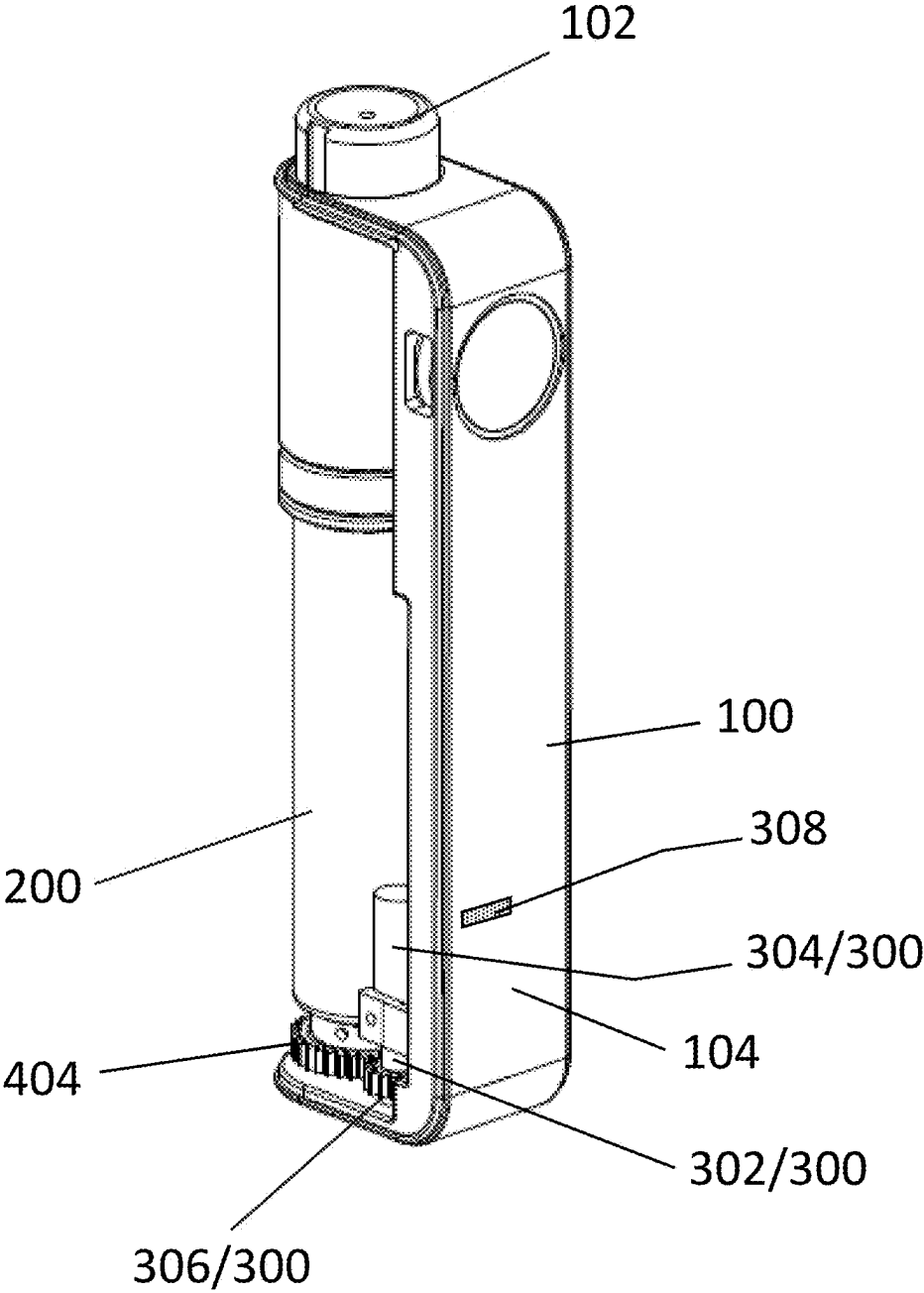


FIG. 3

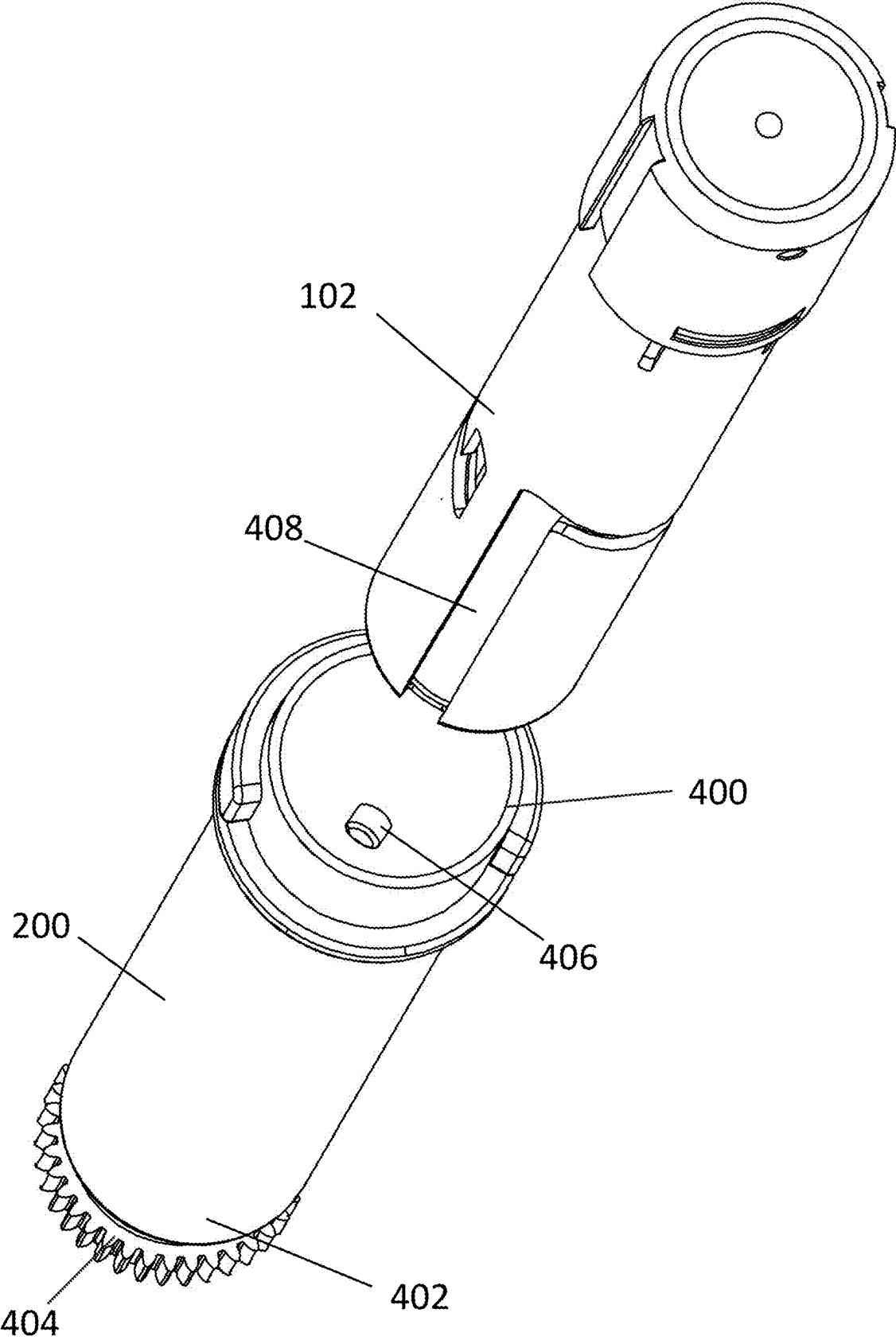


FIG. 4

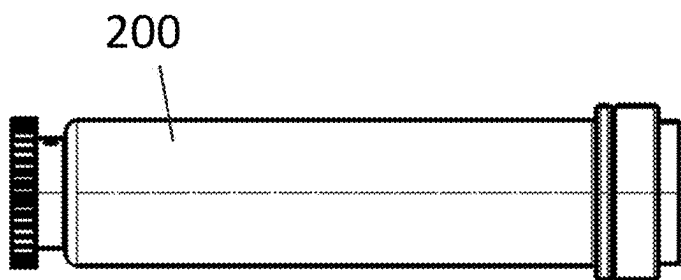


FIG. 5A

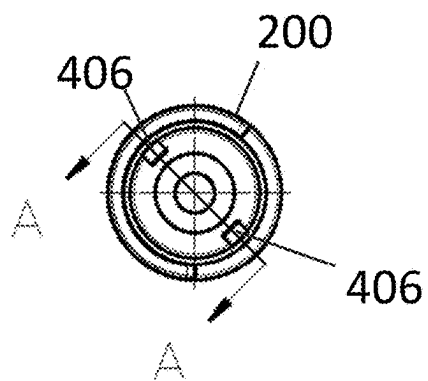


FIG. 5B

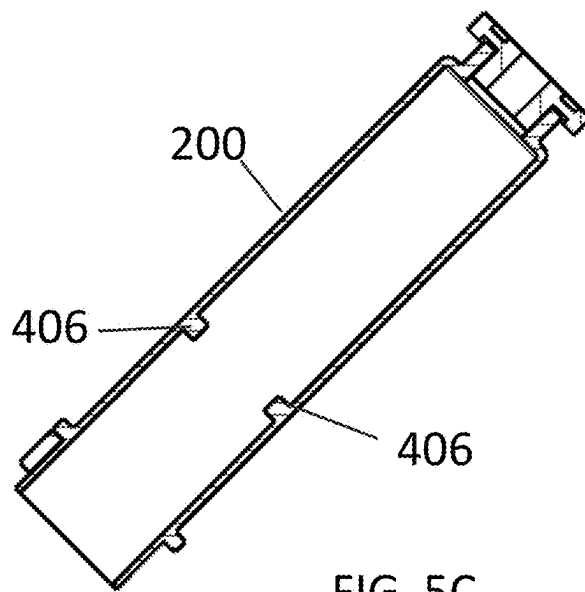


FIG. 5C

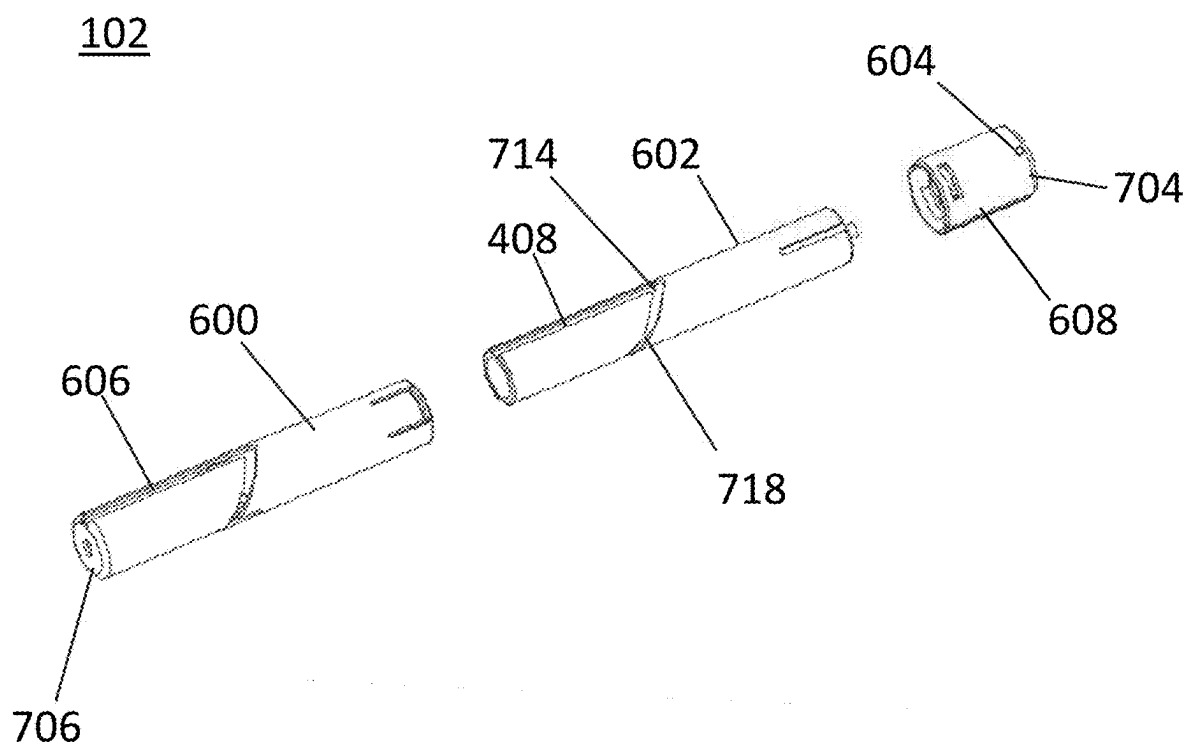
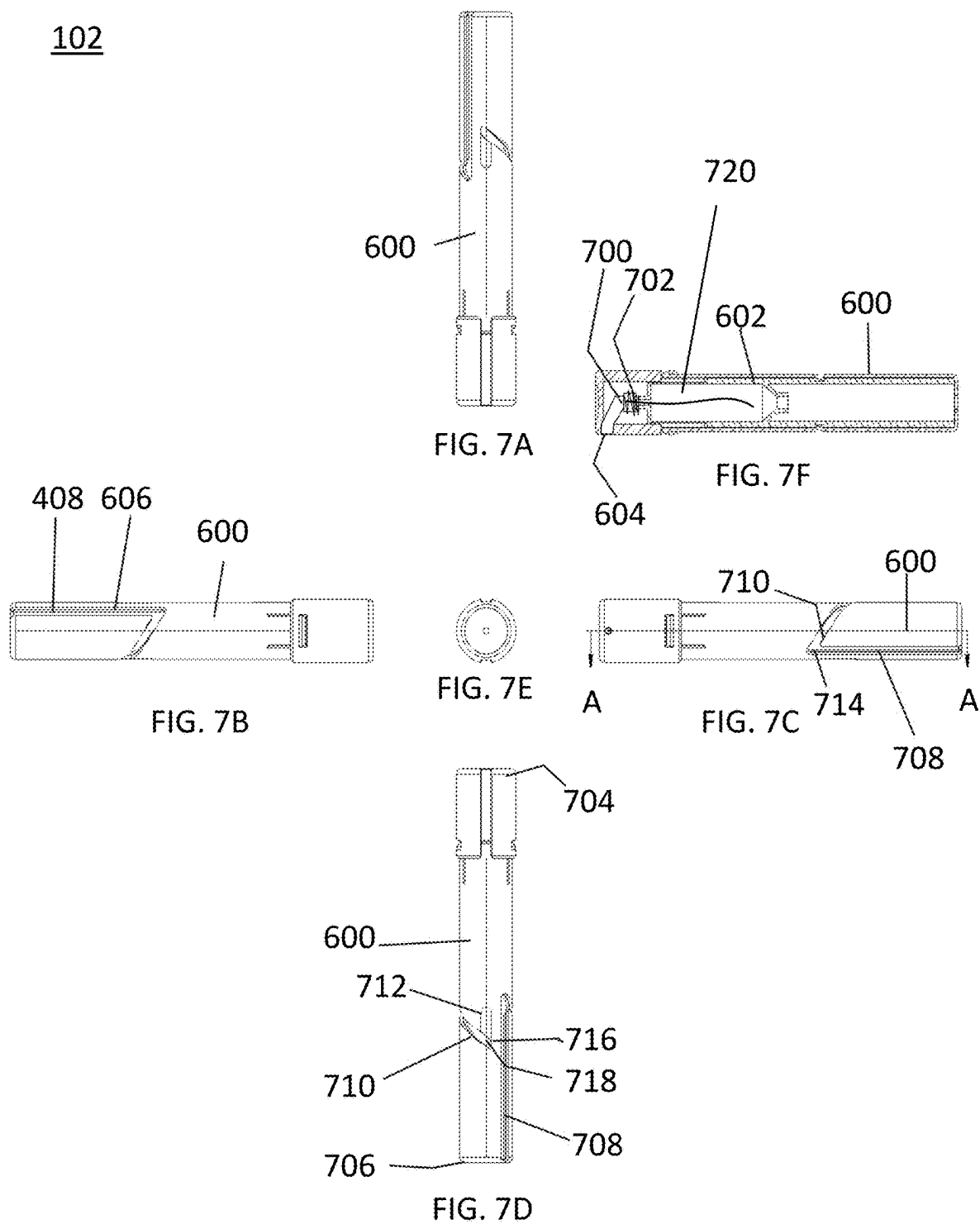


FIG. 6

102



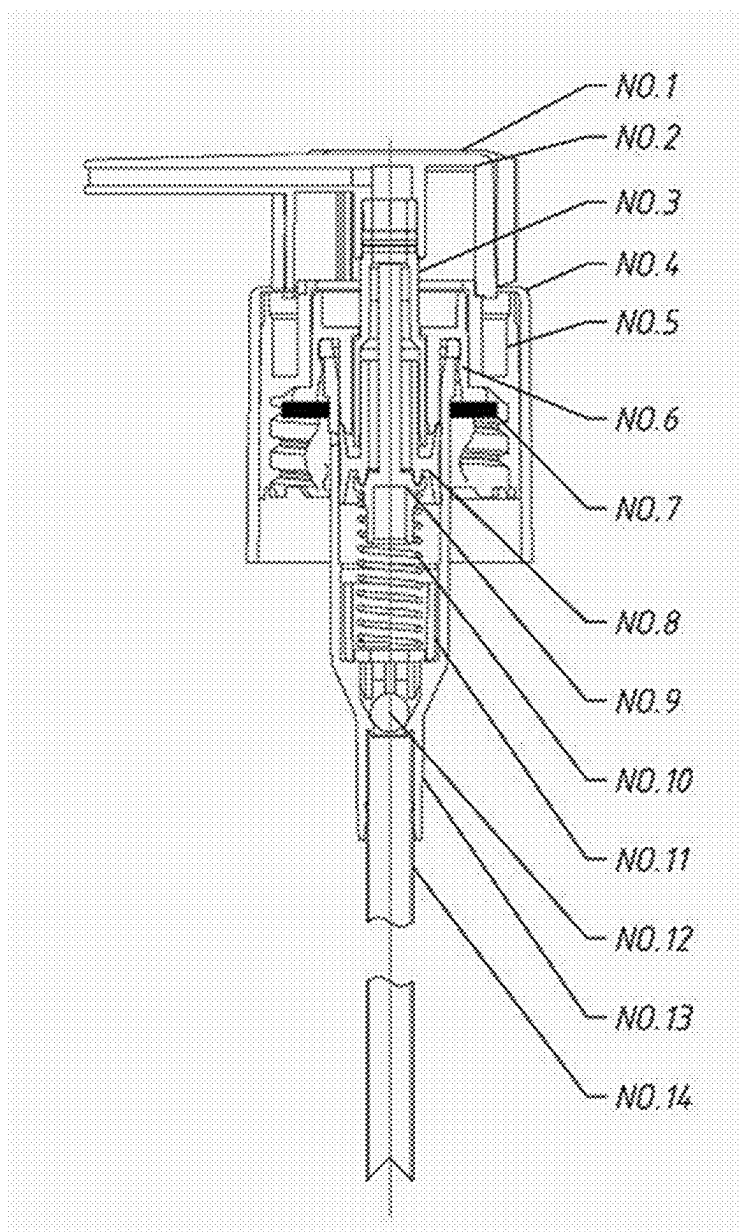


FIG. 7G
(Prior Art)

102

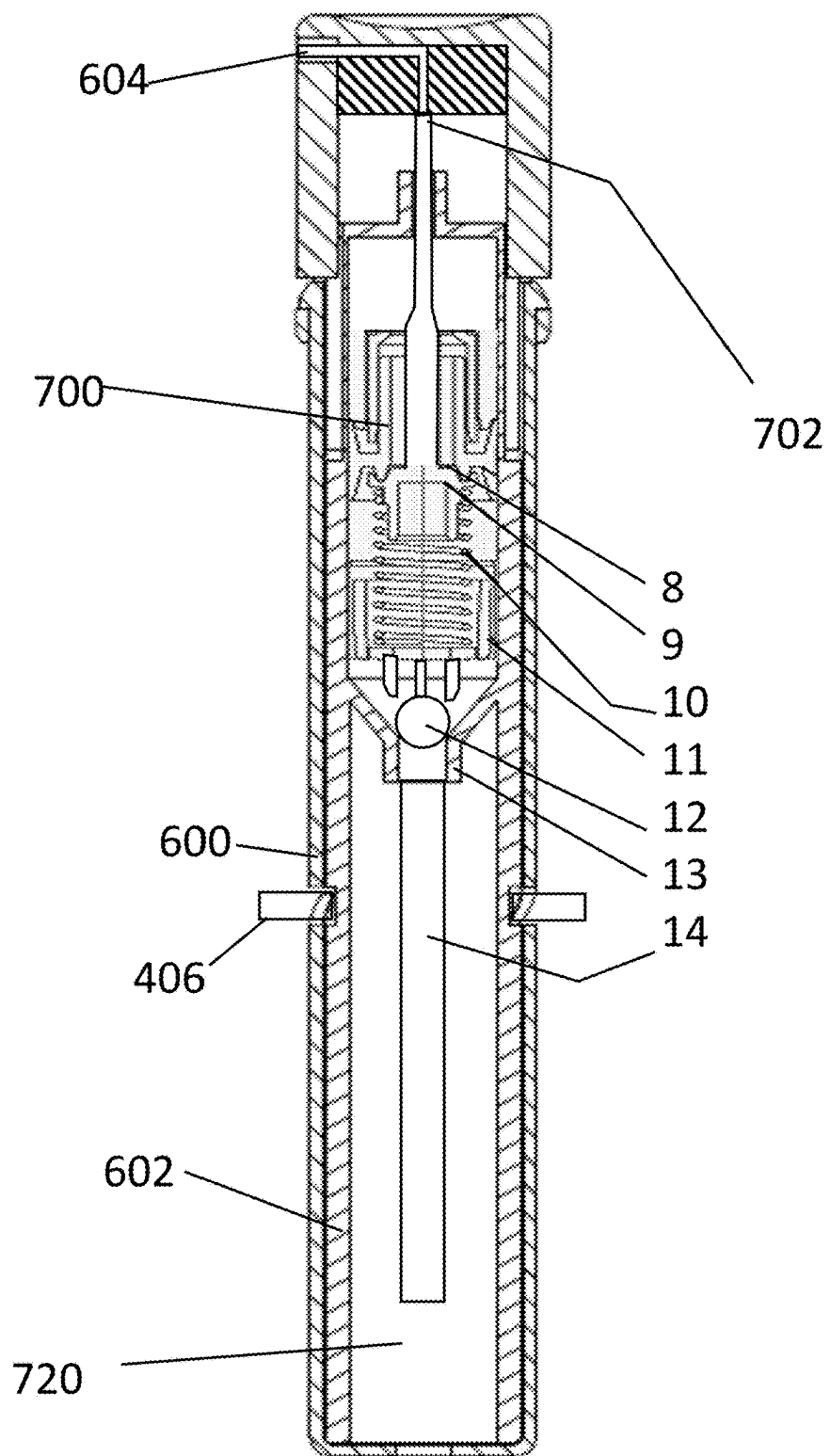


FIG. 7H

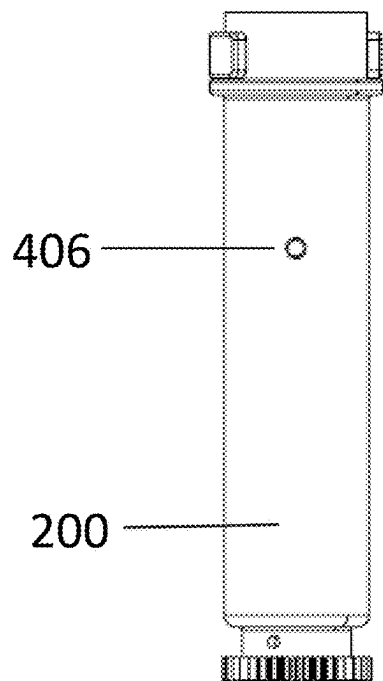
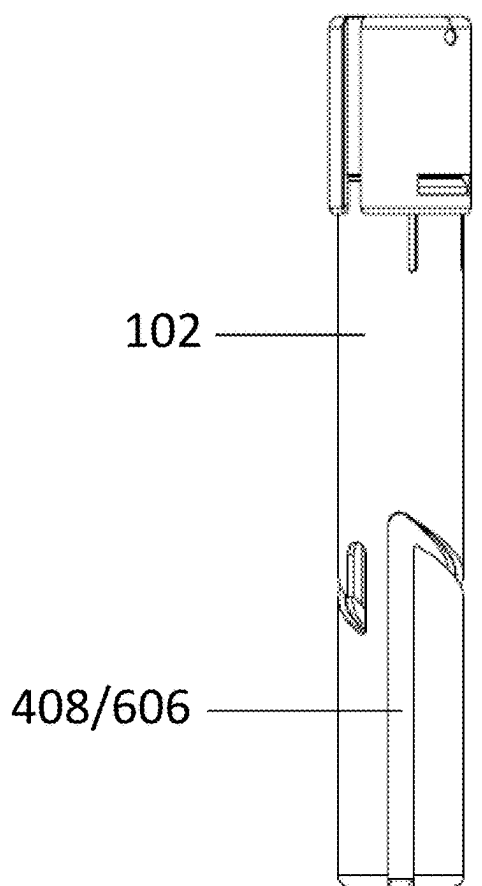


FIG. 8A

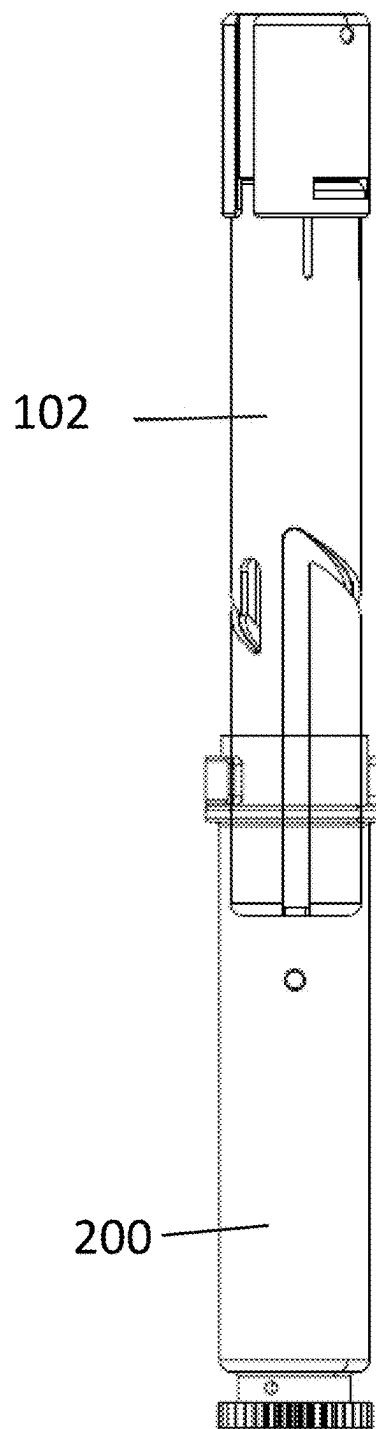


FIG. 8B

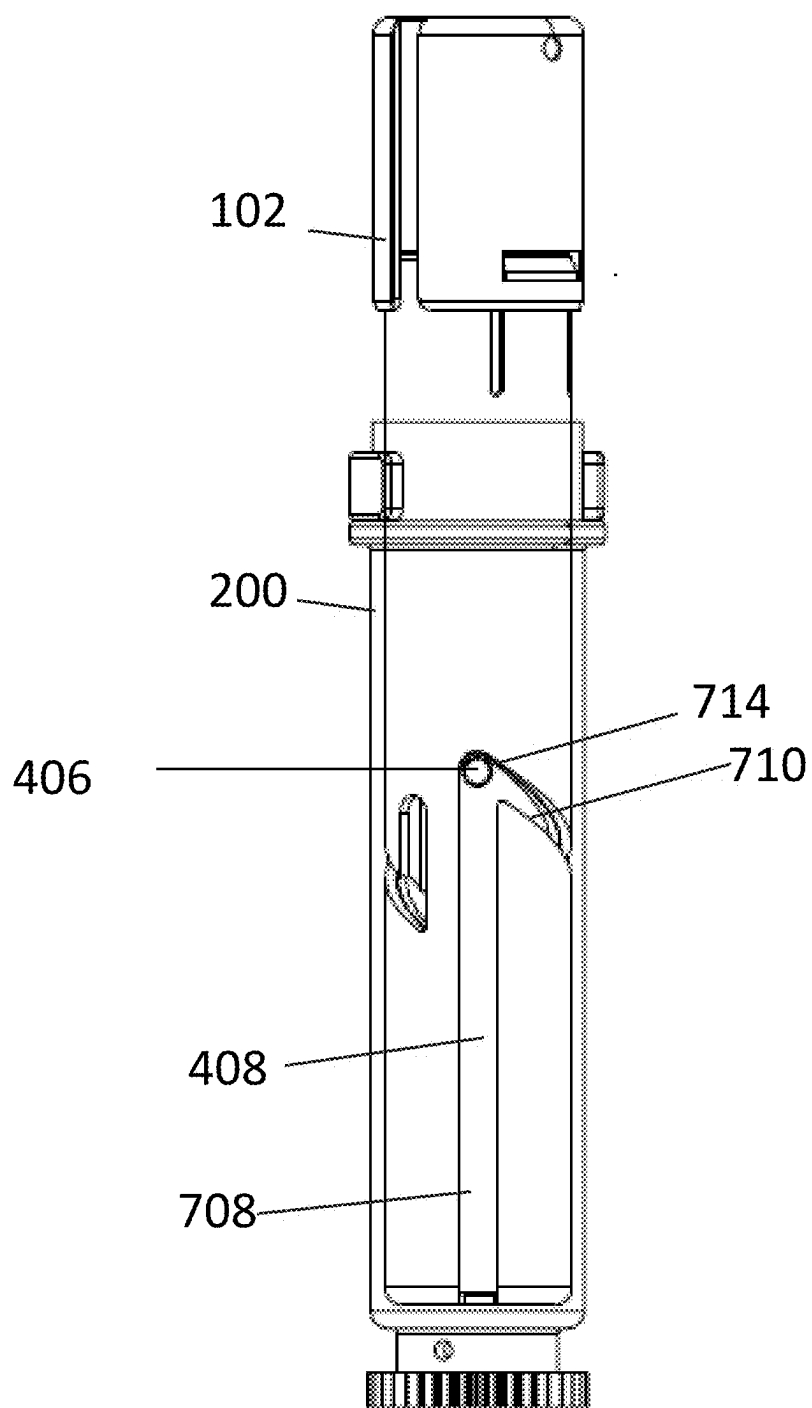
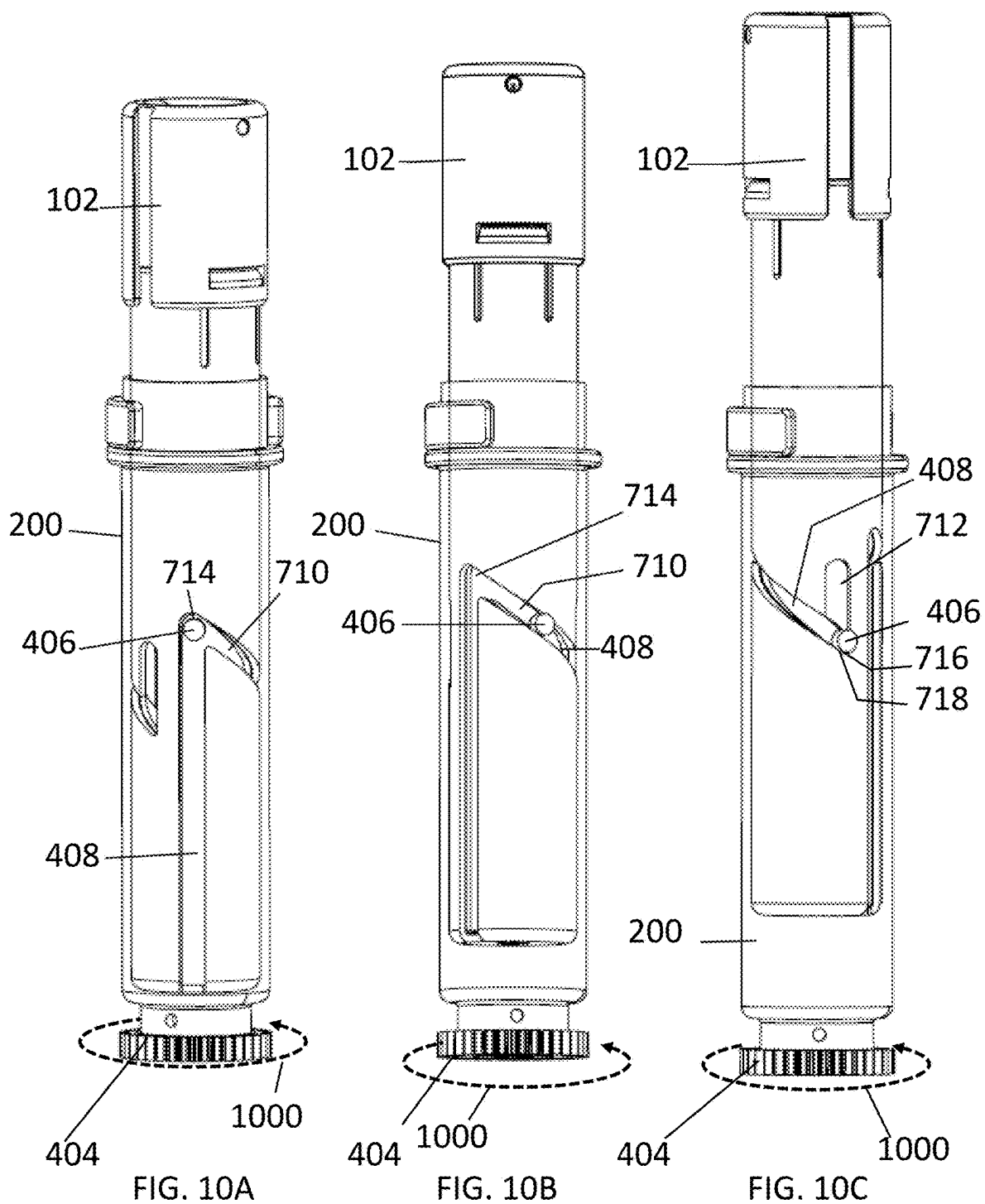
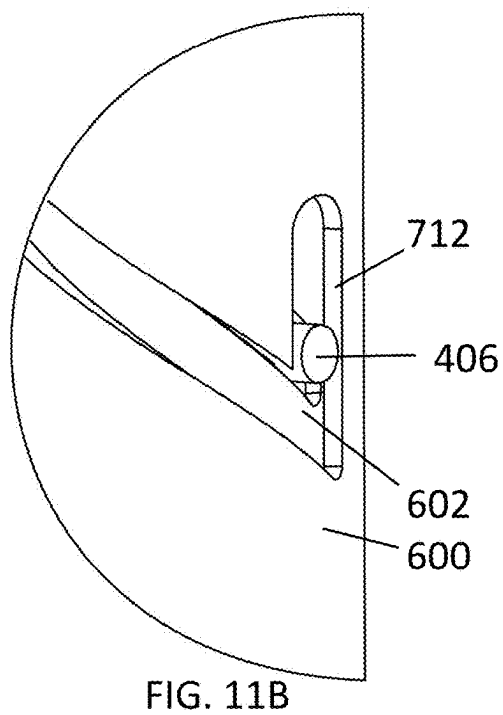
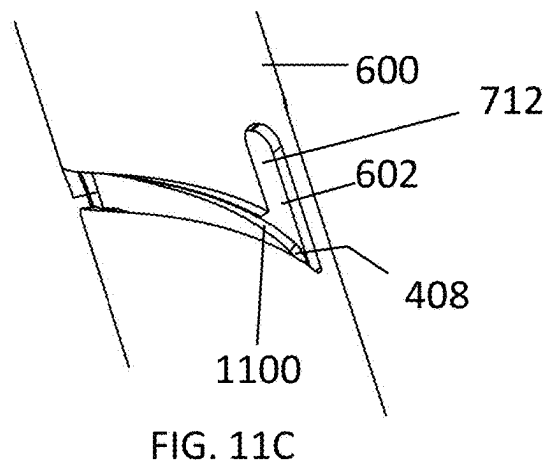
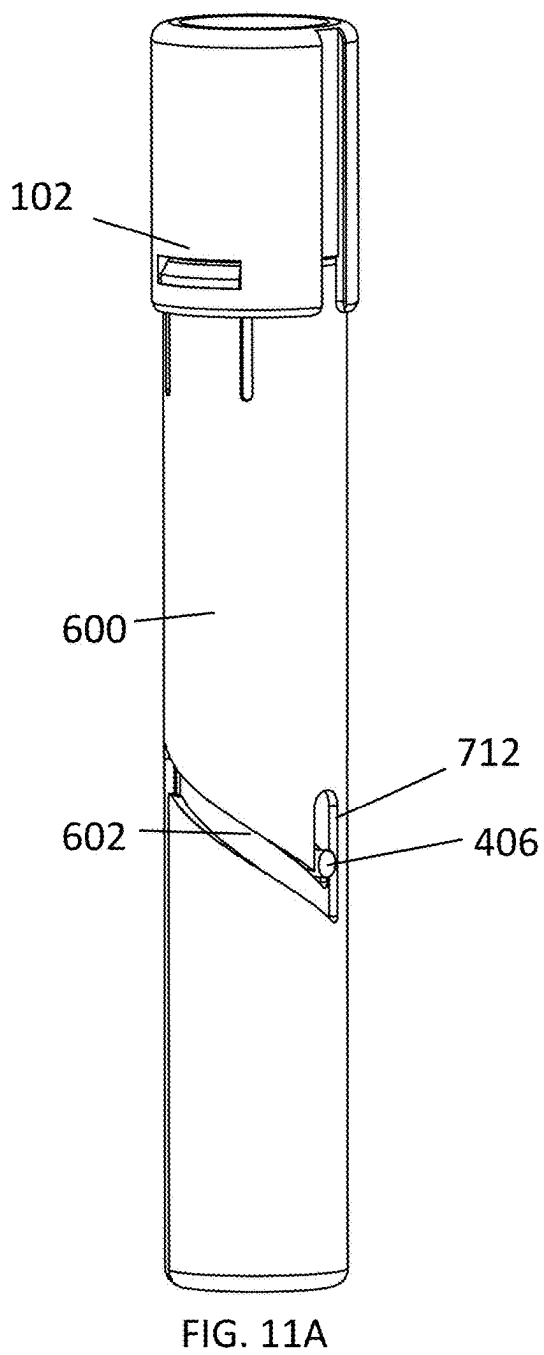
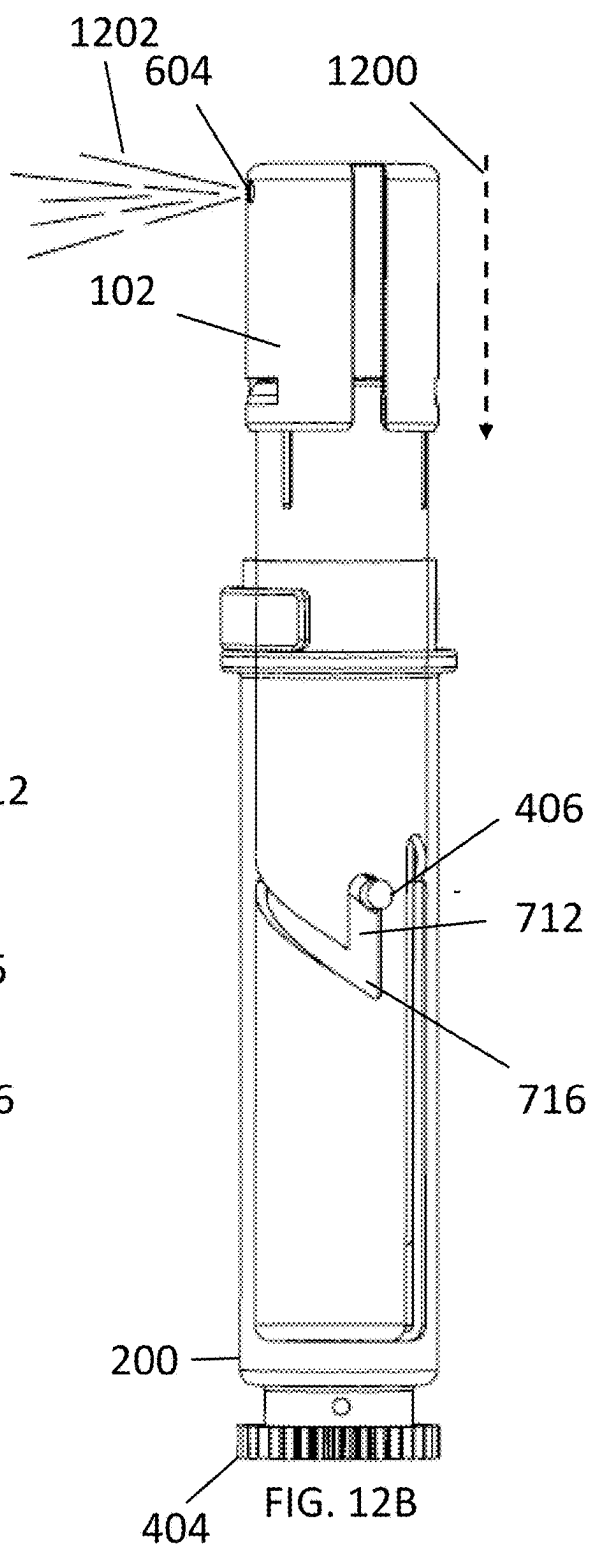
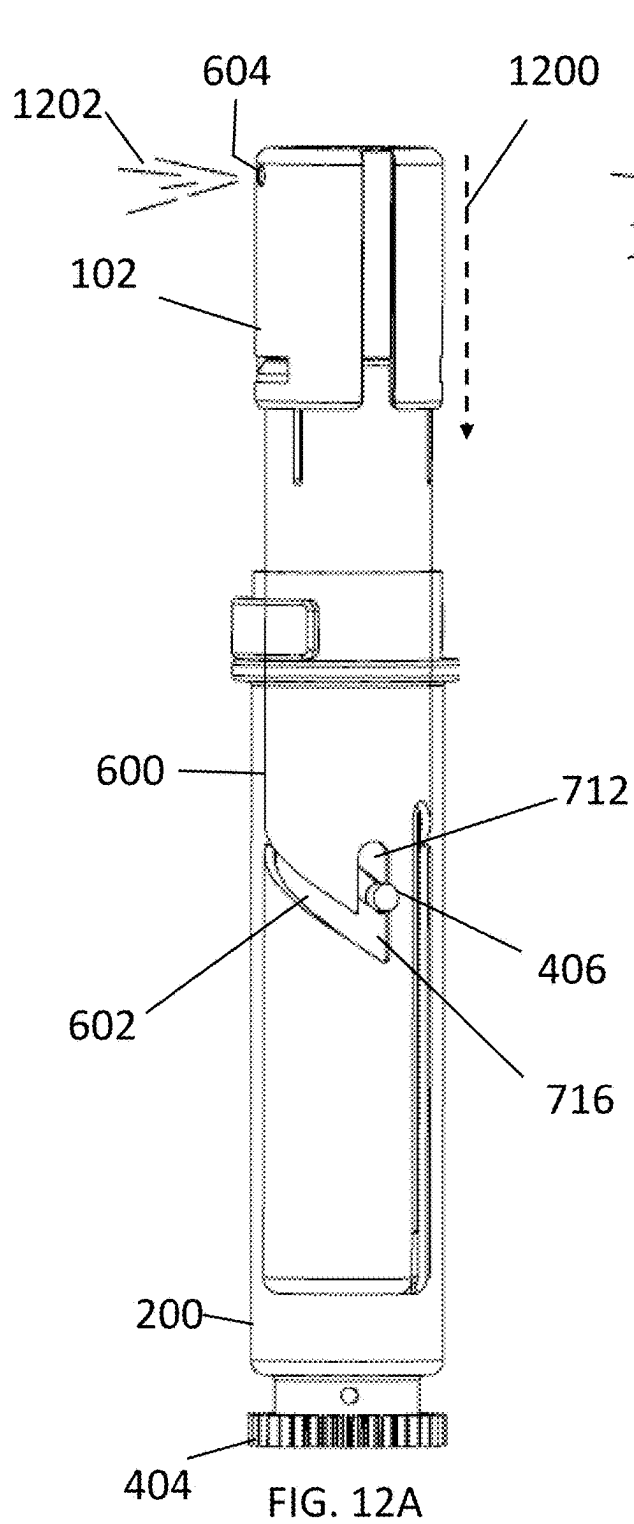


FIG. 9







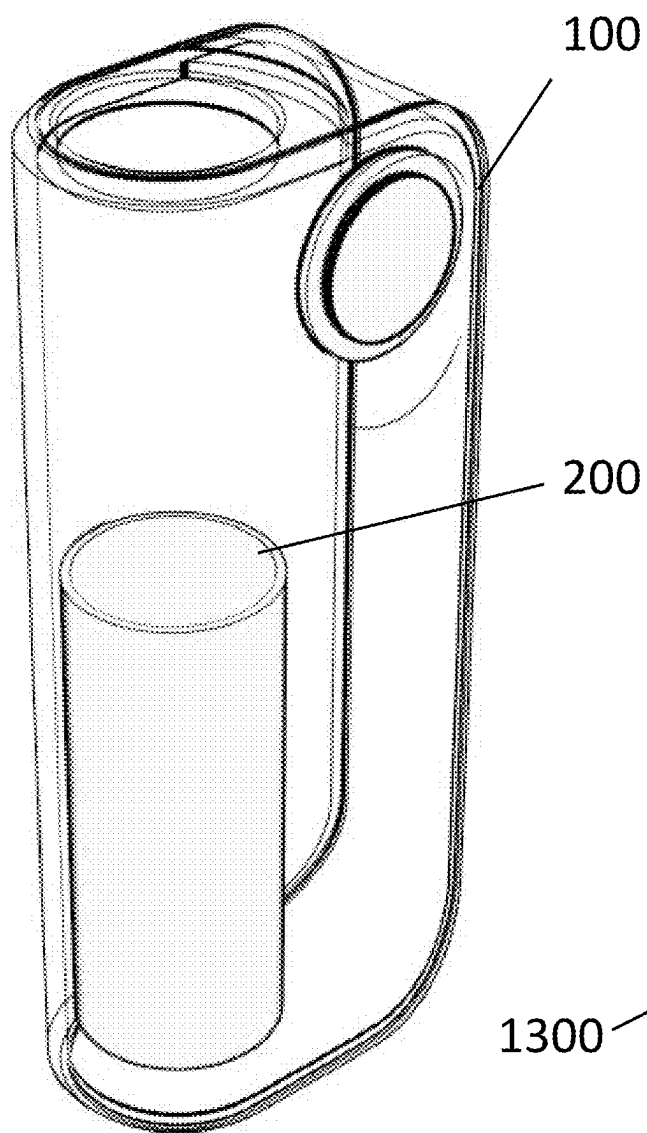


FIG. 13A

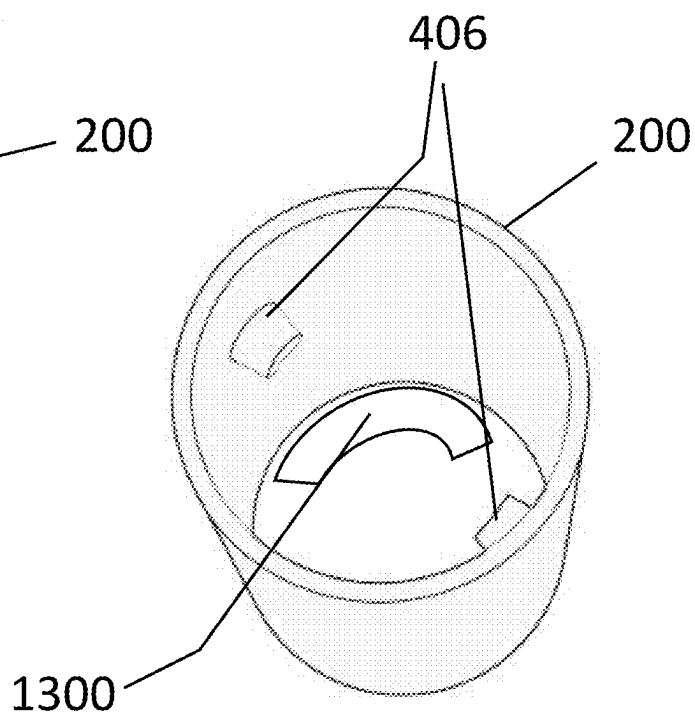
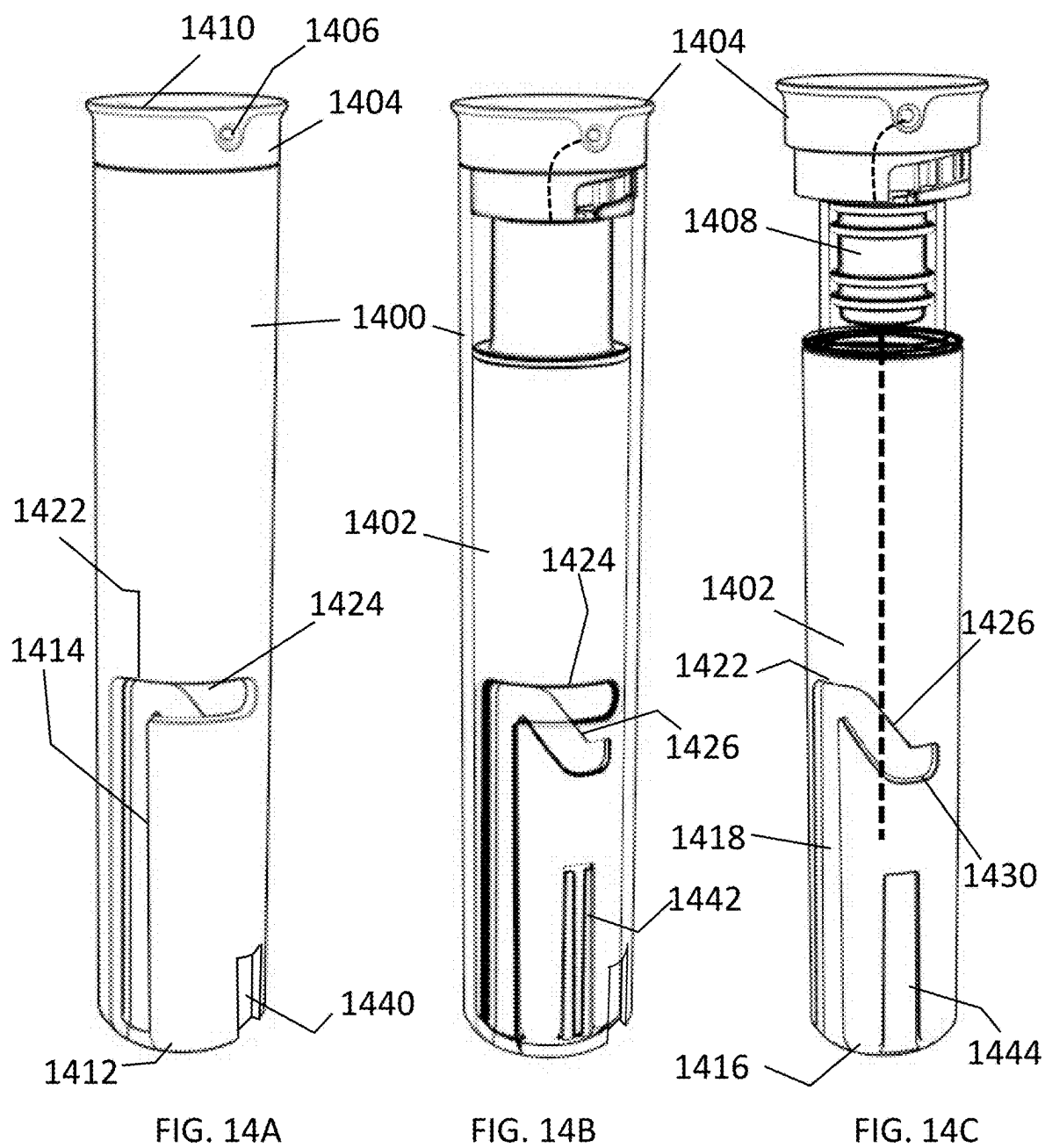


FIG. 13B

101



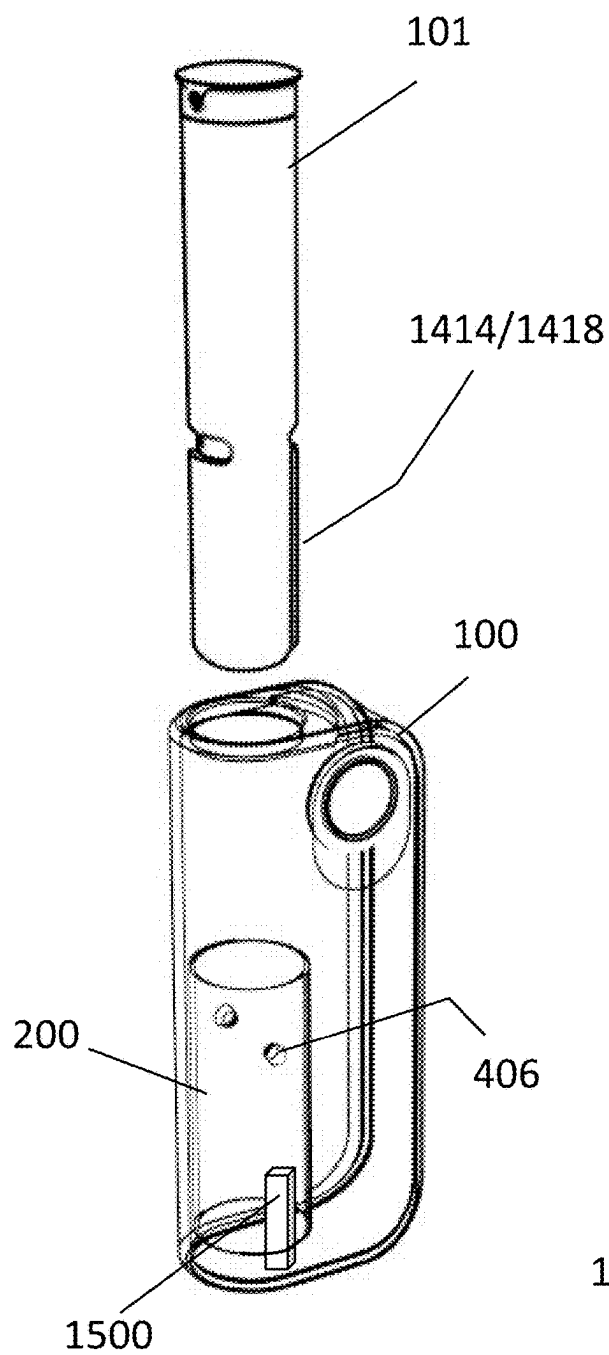


FIG. 15A

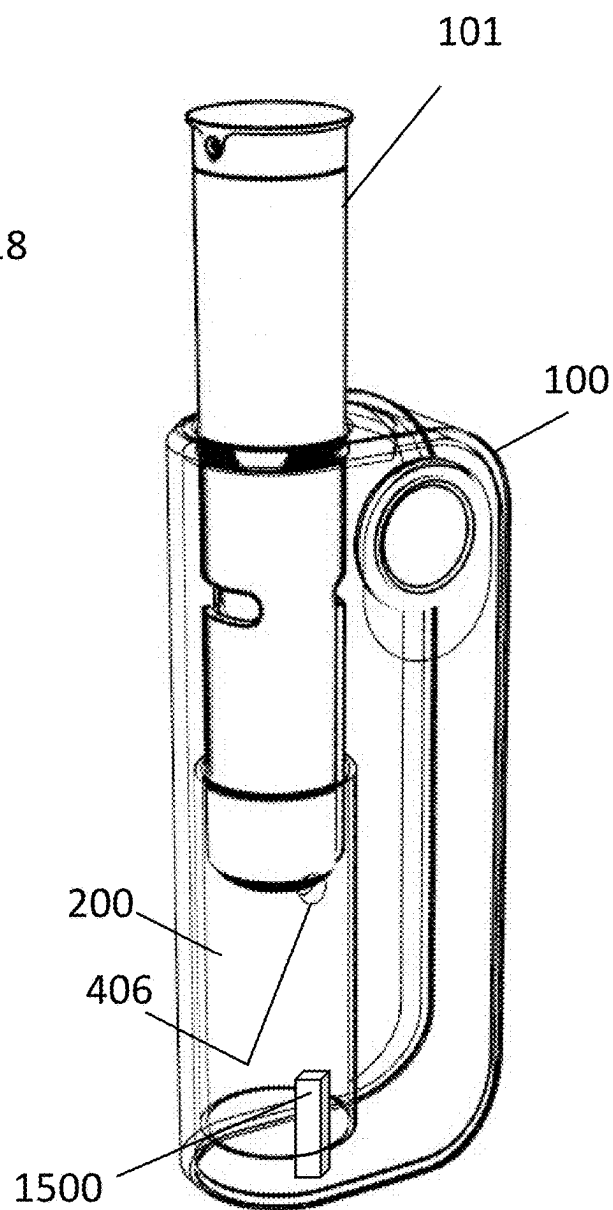


FIG. 15B

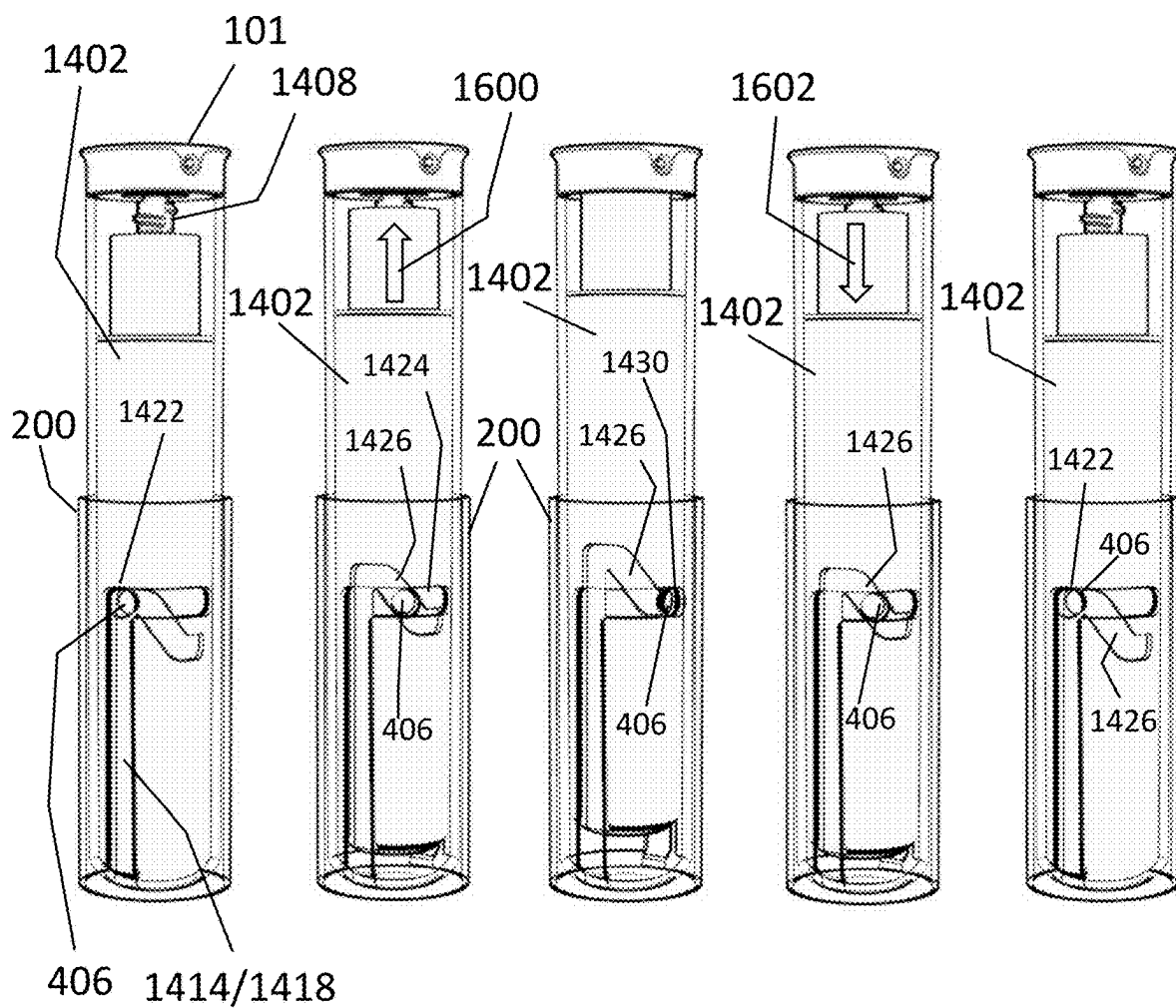


FIG. 16A

FIG. 16B

FIG. 16C

FIG. 16D

FIG. 16E

MEDICATION CARTRIDGE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a Continuation-in-Part application of U.S. Serial No. U.S. Ser. No. 18/600,198, filed Mar. 8, 2024, the entirety of which is incorporated herein by reference.

[0002] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/600,402, filed Mar. 8, 2024, the entirety of which is incorporated herein by reference.

[0003] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/601,329, filed Mar. 11, 2024, the entirety of which is incorporated herein by reference.

[0004] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/655,722, filed May 6, 2024, which is a Continuation application of U.S. Ser. No. 18/606,136, filed Mar. 15, 2024, now issued as U.S. Pat. No. 12,106,836, the entirety of which are incorporated herein by reference.

[0005] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/606,151, filed Mar. 15, 2024, the entirety of which is incorporated herein by reference.

[0006] The present application is ALSO a Continuation-in-Part application of U.S. Ser. No. 18/606,173, filed Mar. 15, 2024, the entirety of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) Field of Invention

[0007] The present invention relates to a medication dispensing system and, more particularly, to a mechanized medication cartridge and associated handheld oral medication dispensing device for actuating the medication cartridge.

(2) Description of Related Art

[0008] The field of medication treatment has grown increasingly complex in recent years, with patients frequently needing to manage and administer multiple types of medication. Under such circumstances, it is vital to provide reliable, accessible, and simple methodologies to patients for dispensing their medications. Many existing dispensing devices, however, tend to be cumbersome, confusing, or inefficient. Often, they rely on patients having to manually change their medication reminders or products, which can involve complicated steps or create potential for mistakes, like grabbing the wrong medication or dispensing an incorrect dosage. Furthermore, mistakes may potentially have serious health implications. Hence, it's vital to design an easy-to-use and fail-safe medication dispensing device.

[0009] Further, existing oral medications are typically taken as tablets or syrups. Such medication delivery items may be functional, but often take time for efficacy due to digestive and other body related functions. Additionally, such medication delivery techniques are inconvenient to carry on one's person or inconspicuously use. Thus, a need exists for a unique oral medication dispensing device that provides an alternative to tablet and syrup-type medications.

[0010] One embodiment disclosed herein pertains to a handheld oral medication dispensing device. To provide for ease of use and reliability, the dispensing device is formed with a mechanized pump. The purpose of this design is not only to make medication administration simpler and more efficient but also to mitigate potential errors between users. The innovative design of a mechanized pump in the dispensing device in conjunction with a keyed connector and the configuration of the cartridges can significantly aid patients to manage and administer their medication effectively and safely.

SUMMARY OF INVENTION

[0011] The present disclosure is directed to a medication cartridge that is operable with an associated dispensing device. The medication cartridge comprises a shell having a dispensing aperture and a vial slidably positioned within the shell, the vial having a pump mechanism with an outlet aligned with the dispensing aperture. The vial is operable for rising within the shell to activate the pump mechanism and dispense medicine from the vial and dispensing aperture.

[0012] In another aspect, the shell includes one or more rising slots passing through the shell and rising from a bottom end of the shell toward an apex.

[0013] In another aspect, the vial includes one or more rising channels that align with the one or more rising slots and rise from a bottom end of the vial toward the apex.

[0014] In another aspect, the shell includes one or more lateral slots that traverse laterally away from the apex.

[0015] In yet another aspect, the vial includes downward descending channels that descend downward at an angle from the apex.

[0016] In yet another aspect, the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.

[0017] Further, the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.

[0018] In yet another aspect, the vial is slidable within the shell between a stored position and a full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.

[0019] Additionally, the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.

[0020] Finally, as can be appreciated by one in the art, the present invention also comprises a method for forming and using the invention as described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The objects, features and advantages of the present invention will be apparent from the following detailed descriptions of the various aspects of the invention in conjunction with reference to the following drawings, where:

[0022] FIG. 1A is an illustration of a first medication cartridge according to a first embodiment;

[0023] FIG. 1B is an illustration of a second medication cartridge according to a second embodiment;

[0024] FIG. 1C shows an exemplary illustration a hand-held medication dispensing device and the first medication cartridge according to the first embodiment;

[0025] FIG. 2A shows an interior-view of the medication dispensing device, depicting a cartridge carrier;

[0026] FIG. 2B shows an interior-view of the medication dispensing device, depicting the first medication cartridge inserted into the cartridge carrier;

[0027] FIG. 3 is an interior-view illustration of the medication dispensing device, depicting an activator mechanism as operably connected with the cartridge carrier;

[0028] FIG. 4 shows an exemplary illustration of the cartridge carrier and first medication cartridge;

[0029] FIG. 5A shows a side-view illustration of the first medication cartridge;

[0030] FIG. 5B shows a top-view illustration of the first medication cartridge;

[0031] FIG. 5C is a cross-sectional, side-view illustration of the first medication cartridge, taken from line A-A of FIG. 5B;

[0032] FIG. 6 shows an exploded-view illustration of the first medication cartridge;

[0033] FIG. 7A shows a front-view illustration of the first medication cartridge;

[0034] FIG. 7B shows a right, side-view illustration of the first medication cartridge;

[0035] FIG. 7C shows a left, side-view illustration of the first medication cartridge;

[0036] FIG. 7D shows a rear-view illustration of the first medication cartridge;

[0037] FIG. 7E shows a top-view illustration of the first medication cartridge;

[0038] FIG. 7F is a cross-sectional, side-view illustration of the first medication cartridge, taken from line A-A of FIG. 7C;

[0039] FIG. 7G is an illustration of a lotion pump assembly of the prior art;

[0040] FIG. 7H is an illustration of an example pump assembly as applied to the first medication cartridge in accordance with various embodiments of the present invention;

[0041] FIG. 8A shows an exemplary illustration of the cartridge carrier and the first medication cartridge;

[0042] FIG. 8B shows an exemplary illustration of the cartridge carrier and the first medication cartridge, depicting the first medication cartridge as being inserted into the cartridge carrier;

[0043] FIG. 9 shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier;

[0044] FIG. 10A shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the first medication cartridge in a stored position;

[0045] FIG. 10B shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier rotating to move the first medication cartridge up and toward the dispensing position;

[0046] FIG. 10C shows an exemplary illustration of the first medication cartridge inserted into the cartridge carrier, depicting the cartridge carrier as rotated to move the first medication cartridge up and into the dispensing position;

[0047] FIG. 11A shows an exemplary illustration of the first medication cartridge, depicting the vial sliding within the shell;

[0048] FIG. 11B shows close-up view of Detail A, taken from FIG. 11A;

[0049] FIG. 11C shows an exemplary illustration of the first medication cartridge, depicting the vial within the shell;

[0050] FIG. 12A shows an exemplary illustration of the first medication cartridge within the cartridge carrier, depicting the first medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the first medication cartridge;

[0051] FIG. 12B shows an exemplary illustration of the first medication cartridge within the cartridge carrier, depicting the first medication cartridge in the dispensing position and being pressed downward to activate the pump assembly and dispense medication from the first medication cartridge;

[0052] FIG. 13A is an interior-view illustration of the medication dispensing device;

[0053] FIG. 13B is an interior-view illustration of the cartridge carrier;

[0054] FIG. 14A is an exterior-view illustration of the second medication cartridge;

[0055] FIG. 14B is an interior-view illustration of the second medication cartridge;

[0056] FIG. 14C is an interior-view illustration of the second medication cartridge, depicting a cartridge shell removed for illustrative purposes to depict the medication vial therein;

[0057] FIG. 15A is an illustration on the second medication cartridge and the handheld medication dispensing device;

[0058] FIG. 15B is an illustration depicting the second medication cartridge being inserted into the handheld medication dispensing device;

[0059] FIG. 16A is an illustration depicting the second medication cartridge in a stored position;

[0060] FIG. 16B is an illustration depicting the second medication cartridge being actuated in an intermediate position to raise the vial and pump medicine from the vial;

[0061] FIG. 16C is an illustration depicting the second medication cartridge being actuated in a full pump position, with the vial completed raised to pump medicine from the vial;

[0062] FIG. 16D is an illustration depicting the second medication cartridge in an intermediate position as it returns to a stored position and lowers the vial; and

[0063] FIG. 16E is an illustration depicting the second medication cartridge in the stored position.

DETAILED DESCRIPTION

[0064] The present invention relates to a medication dispensing system and, more particularly, to a mechanized medication cartridge and associated handheld oral medication dispensing device for actuating the medication cartridge. The following description is presented to enable one of ordinary skill in the art to make and use the invention and to incorporate it in the context of particular applications. Various modifications, as well as a variety of uses in different applications will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to a wide range of embodiments. Thus, the present invention is not intended to be limited to the embodi-

ments presented, but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

[0065] In the following detailed description, numerous specific details are set forth in order to provide a more thorough understanding of the present invention. However, it will be apparent to one skilled in the art that the present invention may be practiced without necessarily being limited to these specific details. In other instances, well-known structures and devices are shown in block diagram form, rather than in detail, in order to avoid obscuring the present invention.

[0066] The reader's attention is directed to all papers and documents which are filed concurrently with this specification and which are open to public inspection with this specification, and the contents of all such papers and documents are incorporated herein by reference. All the features disclosed in this specification, (including any accompanying claims, abstract, and drawings) may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is only one example of a generic series of equivalent or similar features.

[0067] Furthermore, any element in a claim that does not explicitly state "means for" performing a specified function, or "step for" performing a specific function, is not to be interpreted as a "means" or "step" clause as specified in 35 U.S.C. Section 112, Paragraph 6. In particular, the use of "step of" or "act of" in the claims herein is not intended to invoke the provisions of 35 U.S.C. 112, Paragraph 6.

[0068] Please note, if used, the labels left, right, front, back, top, bottom, forward, reverse, clockwise and counter clockwise have been used for convenience purposes only and are not intended to imply any particular fixed direction. Instead, they are used to reflect relative locations and/or directions between various portions of an object.

(1) Description

[0069] As noted above, the present invention relates to a medication dispensing system having a mechanized medication cartridge and, in some aspects, an associated handheld oral medication dispensing device for actuating the medication cartridge. This disclosure provides two example embodiments of the medication cartridge, both of which can be used with the handheld medication dispensing device. FIG. 1A depicts a first embodiment of the medication cartridge 102 while FIG. 1B depicts a second embodiment of the medication cartridge. It should be noted that handheld medication dispensing device as described herein can be programed to work with either medication cartridge, as well as other designs as can be appreciated by those skilled in the art. For clarity, FIGS. 1C through 12B are described and illustrated with respect to the first embodiment of the medication cartridge, while FIGS. 13A through 16E are described and illustrated with respect to the second embodiment of the medication cartridge.

[0070] As shown in FIG. 1C, the handheld medication dispensing device 100 is designed primarily to securely store and dispense medication from a medication cartridge 102. The device 100 comprises a housing 104, principally structured to protect the critical internal components while offering ergonomic suitability for handheld operation. The housing 104 exterior is designed with both convenience and comfort in mind, allowing a user to hold the device 100 and

access the medication with ease. In an exemplary embodiment, the housing 104 may be configured in a variety of shapes, sizes, and materials to accommodate different user preferences and medication types.

[0071] In some aspects the device 100 includes an authentication component 106 that is designed to authenticate a user's identification in various manners and, upon authentication, allow for operation and control of the device 100. The authentication component 106 is any suitable mechanism, device, system, etc., that allows for user authentication, non-limiting examples of which include a fingerprint reader built into the device 100 or facial recognition via a camera and app, etc. that communicates with the device upon authentication to notify the device 100 of authentication and allow for operation of the device 100, or any other known authentication method or device. The authentication component 106 is set up to permit the authenticated or designated user access to the operation of the said device 100, thereby ensuring a highly secure and personal way of dispensing medication. In one embodiment and as illustrated, the authentication component 106 includes a fingerprint reader and all associated components as necessary to allow for authentication with such a reader, including an integrated circuit, power source (battery), etc. For example, in a first use, the authentication component 106 can be configured to recognize the user and, thereafter, only allow access to the recognized user to cause the device 100 to operate as intended. Such a setup can be configured in any suitable manner as understood by those skilled in the art. As non-limiting example, a first use may require that a user turn on the device 100, which provides indicia (e.g., via a light, digital screen, etc.) to notify the user to place their finger on the fingerprint reader. The fingerprint reader then reads and stores the fingerprint of the designated user. Thereafter, the activator mechanism can only be activated by the designated user upon placement of the finger on the fingerprint reader. In another example, software or a phone application can be used to wirelessly communicate (e.g., via Wi-Fi, Bluetooth, etc.) with the device 100 to program the device 100 to only recognize the designated user via the authentication component 106. Thus, in some aspects, the device 100 includes all necessary components to allow it to wirelessly communicate with an external device, non-limiting examples of which include Wi-Fi transceivers/components, Bluetooth components/gear, etc. In an alternative embodiment, the device 100 can simply be turned off and on and includes a button or other similar feature to cause the device 100 to operate and actuate the activator mechanism after user authentication, as described in further detail below.

[0072] In another embodiment and in the case of facial recognition via a camera and app, etc. on a mobile phone that communicates with the device 100, the authentication component 106 includes components housed within device 100 to allow for remote authentication, including the programming on an integrated circuit or similar device, along with the wireless communication components necessary to communication with the mobile phone (e.g., Bluetooth, WiFi transceivers, etc.) to allow for authentication and operation of the device 100. The software or application that is downloaded onto the mobile phone or device includes the programming and any other necessary information to allow for user authentication and cause the authentication to be transmitted and received by the authentication component 106.

[0073] In one aspect, the authentication component 106 is a biometric fingerprint reader linked to a mobile app, which can be used to prevent unauthorized medication dispensing, requiring user authorization via the app for initialization, and connecting to caregivers and personal networks to ensure medication adherence. The design of the device 100, accessories, and app is centered around the need for an easier and better way to receive the right dose, at the right time, in the right way and frequency. In one aspect, controlled via the app, the handheld device 100 can alert users to their medication schedule, while also notifying caregivers and family members about adherence. For restricted medications, the device 100 and associated app can restrict dosing until authorized by the prescribing physician. Tampering with the medication cartridge 102 triggers alerts to caregivers, pharmacists, and physicians, ensuring safe usage, especially for restricted medications by alerting through the app that the medication has not been taken as prescribed. In another aspect, the device 100 and medication cartridge 102 are formed to include any necessary components that allow the device 100 to recognize a specific medication cartridge 102. As a non-limiting example, each medication cartridge 102 can be formed to include a Radio Frequency Identification (RFID) tag, while the device 100 includes an RFID reader (or other similar technology).

[0074] As noted above, in some aspects, the device 100 is formed to wirelessly communicate with an external device (e.g., mobile phone) and its app to allow for the external device to control any features as applicable to the device 100. In doing so, the dispensing device 100 includes a wireless communication component (depicted as element 202 in FIG. 2) which include all necessary components to allow it to wirelessly communicate with the external device and control the dispensing device 100 to provide the desired operations, non-limiting examples of which include Wi-Fi transceivers/components, Bluetooth components/gear (e.g., a system on a chip (SoC) that includes a transceiver, antenna and control chip), integrated circuit(s), a power source (e.g., battery), and/or any another other electronic components as may be needed to allow for wireless or Bluetooth connectivity and related operations of the device 100. In one aspect and per Bluetooth protocol, when the dispensing device 100 links with the associated mobile external device to communicate, they form an ad hoc mini computer network referred to as a piconet. Within the piconet, the external device, referred to as the master, assumes the control of the network, issuing instructions to the dispensing device 100, which is referred to as a slave to control a variety of features and operations.

[0075] As noted above, several programmable features include controlling user access and operation of the device 100 (and activator mechanism, etc.) based on user authentication, dosing schedules, restricting dosing until authorized by a prescribing physician, etc. As a non-limiting example, after a medication cartridge 102 is inserted into the device 100, it is essentially locked and not allowed to position the medication cartridge 102 into the dispensing position until access is granted. This access can be based on the mobile phone user authentication process which transmits an “access granted” signal (via Bluetooth, etc.) to the device 100 and allows the activator mechanism (described in further detail below) to operate and move the medication cartridge 102 to the dispensing position. Without such an “access granted” signal that is wirelessly received by the

dispensing device 100, the device 100 is locked and will not activate the activator mechanism. As yet another non-limiting example, the dosing schedule as recorded into the software application and external device may specify that the medication is to be dispensed a certain number of times daily. A user would then connect their mobile external device (e.g., mobile phone) via Bluetooth to the dispensing device 100 during the allotted times, at which point an “access granted” signal is provided to the dispensing device 100. The software application as stored in the mobile external device will only provide the “access granted” signal that corresponds to the dosing schedule, while the dispensing device 100 is locked or deactivated during all other times. In other words and as can be appreciated by those skilled in the art, there are a variety of features of the dispensing device 100 that can be operated by providing the wireless communication components that allow for wireless control of the dispensing device 100 via a mobile phone or other external wireless device.

[0076] As shown in the interior views of FIGS. 2A and 2B, integrated within the housing 104 is a cartridge carrier 200. The cartridge carrier 200 functions as a receptacle for the medication cartridge 102. The cartridge carrier 200 includes a keyed connector, most importantly, innovatively designed to align with a specified type of medication cartridge 102. This keyed connector serves as an interlocking structure that carefully guides the positioning and orientation of the cartridge 102, allowing only for the insertion of a compatible medication cartridge 102. The keyed connector along with the cartridge carrier 200 ensure the correct placement and alignment of the medication cartridge 102, which can help eliminate potential misuse or cartridge damage, and fostering a smooth transition between stored and dispensing positions (as described in further detail below).

[0077] As shown in the interior view of FIG. 3, an activator mechanism 300 forms part of the configuration and is integrated within the housing 104. The activator mechanism 300, when engaged or otherwise activated, prompts the cartridge carrier 200 to actuate. The involvement of the activator mechanism 300 ensures a controlled movement of the inserted medication cartridge 102 between at least two positions—a stored position and a dispensing position. Note that FIG. 2A depicts the medication cartridge in a stored position, while FIG. 3 depicts the medication cartridge in a dispensing position.

[0078] In the stored position, as shown in FIG. 2A, the medication is kept secure within the device 100. The position ensures safety by preventing unintentional discharge of medication when not in use. When the activator mechanism 300 is activated, the cartridge carrier 200 rotates to lift up the medication cartridge 102 into the dispensing position. When this dispensing position is selected, the medication cartridge 102 is correctly oriented to dispense the medication through a dispensing aperture (described in further detail below) integrated into the medication cartridge 102.

[0079] The link between the activator mechanism 300 and the cartridge carrier 200 allows for reliable and essential movement control. This means the user can confidently operate the device 100 knowing the medication is dispensed accurately and safely.

[0080] The cartridge carrier 200 is formed in any suitable manner to allow for activation by the activator mechanism 300, which results in transitioning the medication cartridge 102 between the stored and dispensing positions. In an

embodiment and as shown in FIG. 3, the cartridge carrier 200 is rotatable within the housing 104. The cartridge carrier 200 may be formed of materials that possess strength and durability such as, but not limited to, metals, polymers, and/or composites. The size, shape, and dimensions of the cartridge carrier may vary as per the application requirements.

[0081] The housing 104 may be constructed to accommodate the rotatable cartridge carrier 200. The construction of the housing 104 can also encompass different types of materials that provide the necessary strength and durability inclusive of, but not limited to metals, polymers, composites, among others. The interior of the housing 104 is ideally designed to allow the cartridge carrier 200 to rotate through actuation of the activator mechanism 300.

[0082] The activator mechanism 300, ensuring the cartridge carrier's 200 rotation within the housing 104, can be implemented through a variety of means, such as mechanical gears, electric motors, a battery, manually operated mechanisms, integrated circuits, or any combination thereof. This activator mechanism 300 administration permits the user to precisely position the cartridge carrier 200 within the housing 104 between a first position (in which the medication cartridge 102 is inserted into the cartridge carrier 200 and is stored in a stored position) and a second position (in which the medication cartridge 102 is lifted from the housing 104 to a dispensing position). The activator mechanism 300 can then be activated to rotate the cartridge carrier 200 in an opposite direction to return the medication cartridge 102 to the stored position.

[0083] In one aspect, the dispensing device 100 includes a locking mechanism that is configured to selectively lock/unlock the medication cartridge 102 within the cartridge carrier 200. The locking mechanism may involve an interlocking system, a magnetic attachment, or any other secure attachment mechanism. As a non-limiting example, the locking mechanism may be a spring-loaded detent style 'click-in' and 'click-out' mechanism formed at the bottom of the medication cartridge 102 and cartridge carrier 200. In another aspect, the locking mechanism can be an electronically actuated lock (e.g., mechanized tab, etc.) that locks the medication cartridge 102 into the cartridge carrier 200. For example, the exterior surface of the medication cartridge can include a slot that is formed to accommodate a tab that is electronically moved (e.g., turned/slid, etc.) into the slot when locking and, in the alternative, pulled from the slot when released. As yet another non-limiting example, the locking mechanism can include programming such that after the cartridge 102 is inserted into the cartridge carrier 202, the cartridge carrier 200 is rotate partially until alignment features (described in further detail below and depicted as element 406) are positioned in the middle of the second slot (described in further detail below and depicted as element 710), with the cartridge carrier 200 then stopped until it is desired to rotate further to position the medication cartridge 102 in the dispensing position. With the alignment features in the middle of the second slot, the cartridge 102 is effectively locked within the cartridge carrier 200. Thus, as can be appreciated by those skilled in the art, there are several mechanisms that can be employed to selectively lock the medication cartridge 102 into the cartridge carrier 200.

[0084] In another aspect, the inclusion of the locking mechanism may also be included to secure the cartridge carrier 200 at any desired rotational position, thus prevent-

ing unintentional and unwanted movement when the cartridge carrier 200 is set in a particular position. This locking mechanism can be integrated into the cartridge carrier 200, the housing 104, the activator mechanism 300, or combinations of these components. This rotatable cartridge carrier 200 system installed within a housing 104 assembly may have wide ranging applications, such as, for instance, accommodating various cartridges for providing users with quick and easy access to different medications and dosages, multiple configurations, interchangeability of components, and more. Thus, the present one embodiment discloses a rotatable cartridge carrier 200 system, promoting ease of use, enhancing performance, providing efficient operation, flexibility, improved control and precision, leading to a more efficient and user-friendly functionality.

[0085] In one embodiment and as shown in FIG. 4, the cartridge carrier 200 includes both a proximal end 400 and a distal end 402. The proximal end 400 of this unit is open, and it is specifically structured to receive the medication cartridge 102. This can be any kind of medication cartridge 102, and the open design simplifies the process of interchanging cartridges of differing medications, volumes, or other specifications, thus making it broadly applicable and versatile.

[0086] The distal end 402 of the cartridge carrier 200 includes a key component of this one embodiment—a gear wheel 404. This gear wheel 404 is not a mere inclusion but is operably connected to the activator mechanism 300. The gear wheel 404 and activator mechanism 300 are conceptually designed to work in harmony. In operation, when the user deploys the activator mechanism 300, the included gear wheel 404 is rotated, thereby rotating the cartridge carrier 200.

[0087] The gear wheel 404 is a significant element in the functioning of this device 100 mainly due to its interaction with the activator mechanism 300. The gear wheel 404, upon the interaction with activator mechanism 300, contributes to the precision with which the cartridge carrier 200 is rotated to position the medication cartridge 102 between the stored and dispensing positions. Its size, teeth count, and interaction with other gears if connected, contribute to its function.

[0088] Moreover, the position of the gear wheel 404 at the distal end 402 of the cartridge carrier 200 may also assist in a user-friendly, ergonomic design. The gear wheel's 404 location ensures a seamless connection between the mechanical parts of the activator mechanism 300 and the carrier cartridge 200, augmenting the device's 100 overall performance.

[0089] As noted above and referring again to FIG. 3, the device 100 includes an activator mechanism 300 to selectively turn the gear wheel 404 and, by extension, rotate the cartridge carrier 200. The activator mechanism 300 includes any components as necessary to rotate the gear wheel 404. In one embodiment, the activator mechanism 300 comprises of at least a power source (e.g., battery 304), a motor 302, and one or more gears 306 that are operably connected to the gear wheel 404. The battery 304 in the device 100 forms the power source, capable of providing a stable and reliable supply of power to the other components. The size, capacity, and type of the battery 304 may be customized according to design preferences, device requirements, or various other factors, while being compliant with standard safety and operation guidelines. In some aspects, the battery 304 is encased within the device 100 and intended for device 100

replacement upon depletion of the battery 304, while in other aspects, it is replaceable. In yet other aspects, the battery 304 is rechargeable using a charging port 308 or any other charging means as known to those skilled in the art.

[0090] The motor 302 functions as the primary mechanical component that is responsible for movement or force in the device 100. Upon activation, the motor 302 draws power from the battery and begins to turn to rotate any of the one or more gears and the operably connected gear wheel 404. The type, size, and characteristics of the motor 302 can be adjusted based on the requirements of the device, anticipated load, desired performance factors among other parameters.

[0091] The one or more gears represent a transmission mechanism, operably connected to the motor 302 and the gear wheel 404. Upon the activation of the device 100, the rotational force produced by the motor 302 is transferred to these gears. The gears enable the transfer of power from the motor 302 to the gear wheel 404. Upon receiving the force from the one or more gears of the activator mechanism 300, the gear wheel 404 begins to turn. This turning of the gear wheel 404 then contributes to the rotation of a cartridge carrier 200.

[0092] Referring again to FIG. 4, one embodiment pertains to an enhanced cartridge carrier 200 system, specifically designed with one or more alignment features 406 that protrude into the cartridge carrier 200 for a keyed connection and accurate cartridge 102 positioning, thereby improving the overall functionality of the cartridge 102 and the cartridge carrier 200 system.

[0093] The alignment features 406 of one embodiment are distinguishable protrusions present on the inner surface of the cartridge carrier 200. These alignment features 406 serve to guide the cartridge 102 into the correct position within the cartridge carrier 200 when installing a cartridge 102 into the system. These protruding alignment features 406 also help to prevent misalignment and displacement of the cartridge 102 while it is installed and during operation. Notably and as described below in further detail, the alignment features 406 are critical in transitioning an inserted cartridge 102 between the stored and dispensing position.

[0094] The alignment features 406 are strategically located within the cartridge carrier 200 in a manner to engage with corresponding features or portions on the cartridge 102. When the cartridge 102 is inserted into the carrier 200, the protruding alignment features 406 guide it into the correct position as the features 406 fit into compatible sections (i.e., alignment channels 408) of the cartridge 102. This results in a snug and correct fit of the cartridge 102 within the cartridge carrier 200.

[0095] Furthermore, these alignment features 406 may come in various shapes and sizes and can be constructed using various materials that are suitable for the system and the intended application. They can be rigid to withstand the pressure exerted by the cartridge 102 as it is inserted into the cartridge carrier 200, or flexible to allow for slight variations in cartridge 102 size and shape.

[0096] For further understanding, FIGS. 5A through 5C depict side, top, and cross-sectional views, respectively, of the cartridge carrier 200 and alignment features 406. These features 406, protruding into the interior of the cartridge carrier 200, support the accurate placement of cartridges 102 within the device. Thus, the functionality and efficiency of the overall dispensing device 100 are significantly improved along with consistent cartridge 102 performance ensured by

these alignment features 406. This inventive cartridge carrier 200 system design genuinely adds value to applications where precise alignment of cartridges 102 in their carrier systems is crucial. This one embodiment indeed ensures an innovative update for improving the efficiency of cartridge insertion and alignment in cartridge carrier systems, potentially paving the way for advancements in devices utilizing such systems.

[0097] As noted above and referring again to FIG. 4, the present disclosure also provides a unique medication cartridge 102 that can be inserted within a cartridge carrier, thus providing a novel means to carry, store and administer medication. The essence of this one embodiment lies in the exclusive design of the medication cartridge 102 and the cartridge carrier that is not only easy to operate but also provides secure storage and effortless administration of the medication.

[0098] The medication cartridge 102 is an integral part of the inventive concept and is designed to contain distinct types of medications, be it in liquid, gel, or cream, or other dispensable forms of medication. The cartridge 102 can be made up of pharmaceutically safe materials ensuring no chemical reactions occur when in contact with the medication. The cartridge 102, itself can showcase cylindrical or other geometrical shapes as required by shape of the cartridge carrier 200. The medication cartridge 102 and cartridge carrier 200 partnership provides an ideal solution in various medical environments, such as hospitals, pharmacies, nursing homes, or even domestic settings. It proves to be beneficial where multiple medications should be stored and administered safely and hygienically. It combines innovation with convenience, improving medication storage, and administration methodologies.

[0099] As noted above and as shown in FIG. 6, the cartridge 102 itself is also designed to provide a marked improvement over other medication containers. It is envisioned to include markings or identification systems for easily recognizing the type of medication stored within each cartridge 102. The real innovation lies in the incorporation of one or more channels 408 and aligned slots 606 within the cartridge 102. These are constructed meticulously within the cartridge 102 and serve a distinct function of aligning and receiving one or more alignment features 406. As will be evident below, these channels 408 and slots 606 engage with the aforementioned alignment features 406 to allow for controlled operation of the device 100.

[0100] Further and as shown throughout FIGS. 6 through 7F, the present disclosure provides an improved configuration for a medication cartridge 102, particularly advantageous for medications that require enhanced storage for securely maintaining their efficacy and quality. The innovative design of the medication cartridge 102 includes a robust shell 600 that has a dispensing aperture 604 designed for the release of the medication stored within a medication vial 602.

[0101] The shell 600, formed from long-lasting and reliable material (e.g., plastic, metal, etc.), is shaped to house the vial 602 securely and efficiently. This shell 600 acts as a protective layer for the vial 602 placed inside, while maintaining the required conditions for the preservation of the medication. The dispensing aperture 604 deployed on the shell 600 is engineered in a way to align with the outlet 702 of the vial 602, ensuring precise and mess-free dispensation of the medication.

[0102] Inside the shell 600 is a medication vial 602, which is shaped and positioned in a manner that it can slide without difficulty within the shell 600. This ability of the vial 602 to slide enables convenient load and unload of the medication. The vial 602 includes a pump assembly 700 that is equipped with an outlet 702. The design and positioning of this outlet 702 are such that it aligns perfectly with the dispensing aperture 604 on the shell.

[0103] The pump assembly 700 is any suitable assembly that can be affixed with a container (i.e., the vial 602) to allow for pumping or otherwise dispensing a medication or fluid therefrom, a non-limiting example of which includes a compression or spring-loaded pump, similar to the pump mechanism as used in a standard lotion bottle. For reference, FIG. 7G provides an illustration of an example pump assembly as provided for in the prior art, while FIG. 7H provides a non-limiting example of the pump assembly as modified and applied to the present invention. The pump assembly of the prior art (as shown in FIG. 7G) includes a clamp 1, press head 2, press lever 3, coat 4, screw cap 5, connect cap 6, gasket 7, piston 8, piston seat 9, spring 10, spring seat 11, valve 12, housing 13, and a straw 14. As applied to the present invention (as depicted in FIGS. 7F and 7H), the pump assembly 700 is connected to the vial 602, while the shell 602 serves as the press head that is used to press down the piston 8 to force the fluid through the straw 14 and out of the dispensing aperture 604. In some instances the pump assembly 700 can be screwed onto the vial 602; however, it should be understood that the invention is not intended to be limited thereto as the pump assembly 700 can be attached in any suitable manner between the vial 602 and shell 600 to allow for dispensing fluid from the vial 602 and out of the dispensing aperture 604. Further, it should be understood that the specific pump assembly 700 as depicted in FIG. 7H is provided as a non-limiting example of a suitable pump assembly and that the invention is not intended to be limited thereto as other mechanisms or assemblies can be used to dispense the fluid upon actuation. For example, the pump assembly can be formed as an airless pump, a peristaltic pump, a diaphragm pump, a pressurized canister style pump, or any other configuration in which activation of the assembly causes the fluid to dispense.

[0104] For illustrative purposes in comparison to a lotion bottle, the vial 602 would serve as the container to hold the fluid, while the shell 600 serves as the pump head that can be depressed to pump the fluid, with the various components of the pump assembly 700 positioned therebetween. In operation and when the medication cartridge 102 is moved to the dispensing position, the alignment features 406 operate to hold the vial 602 in place (due to their locking position in the channel terminal as described in further detail below), while a user can selective press the shell 600 downward around the vial 602 to activate the pump assembly 700 and force fluid from the vial 602.

[0105] For example, the vial 602 includes a reservoir 720 in which the medication is stored. The medication can be in a gas or fluid form; however, desirably and in one aspect, the medication is fluid and can be squirted from the medication cartridge 102 (and vial 602) via actuation of the pump assembly 700. As apparent from the description further below, the pump assembly 700 can be easily operated to cause the medication to be dispensed from the vial 602 through its outlet 702. When a force is applied on the pump assembly 700, it triggers the release of the medication inside

of the vial 602, causing the medication to travel from the vial 602, passing through the aligned outlet 702 and dispensing aperture 604, where it is released to the user.

[0106] As noted above, the pump assembly 700 is any suitable assembly, mechanism, or device that can be affixed with the vial 602 to allow for selective dispensing therefrom. As can be appreciated by those skilled in the art, a non-limiting example of such a pump assembly 700 is a spring-loaded system utilizing a stacked one-way valve system to temporarily create a vacuum, allowing fluid to be aspirated from the reservoir 720 and dispensed through the outlet 702 and dispensing aperture 604 upon depression of the shell 600 around the vial 602.

[0107] As noted above, the medication cartridge 102 is generally formed of two components, a shell 600 and a medication vial 602. The vial 602 is sized to be secured within and slide within the shell 600. As shown, the medication cartridge 102 includes a top end 704 and a bottom end 706. Although the shell 600 in FIG. 6 is illustrated as having a cap component 608 at the top end 704, the invention is not intended to be limited thereto as the shell 600 can be formed such that the cap component 608 is integrally formed with the shell 600 as a single unit. In such an aspect, a hole (e.g., sealable hole) can be formed at the bottom end 706 (or top end 704) of the shell 600 to allow for insertion of the vial 602 into the shell 600.

[0108] Notably and of particular importance, the vial 602 includes channels 408 that are aligned with the slots 606 on the shell 600. These channels 408 and slots 606 are built in a manner to accurately align with the alignment features 406 of the cartridge carrier 200. The number and size of these channels 408 and slots 606 can vary depending on the number of alignment features 406 intended to be placed therein. These channels are purposefully built to receive these alignment features, allowing secure and precise positioning. This design ensures that the alignment features do not shift their placement, further guaranteeing the stabilization of the medication within the cartridge.

[0109] For example, one example embodiment includes a structure involving a pair of opposing channels 408 are formed on opposite sides of the vial 602. These channels 408 are typically parallel to each other, though variations can occur. In one aspect, each of the pair of opposing channels 408 originates from the lowermost end or bottom end 706 of the vial 602. The pair of opposing channels 408 rise upward from this point, extending towards the top end 704 of the vial 602. This ascension continues until they reach a point known as the first junction 714. It must be clarified that this first junction 714 is not an arbitrary point but is a strategically designed and located structural feature within the device to precisely position the cartridge 102 between stored and dispensing positions.

[0110] Once these channels 408 have reached the first junction 714, the trajectory shifts. At this point, the channels 408 do not continue with the upward gradient. Instead, they commence a downward traversal toward the bottom end 706. They proceed downwards, but not vertically downwards. This descent is carried out on an angle around the vial 602, intended to enhance function and performance.

[0111] These angled downward channels 408 continue to move in the said direction until they reach an area deemed as the second junction 716. It is at this point that these channels 408 cease to continue. They terminate at this juncture, each channel 408 ending in what is referred to as

a channel terminal **718**. As will be apparent below, this distinctive configuration of the channels **708** enables them to perform their intended purpose more efficiently. It is important to consider the precise and specific architectural design of these channels **408**, including their starting point at the bottom end **706** of the vial **602**, their rise towards the first junction **714**, their angled descent towards the second junction **716**, which with respect to the channels, is the final termination at the channel terminal **718**. These particular design elements reveal a thoughtful consideration of their role within the device and contribute both independently and collectively to the overall functionality and performance of the device or system in question.

[0112] The present disclosure also provides an embodiment which is characterized by a pair of opposing slots **606** in the shell **600** that align with the pair of opposing channels **408**. More specifically, one embodiment is the configuration and interaction of these parts, enabling the pair of opposing alignment features **406** (also referred to herein as protrusions **406**) to pass through the pair of opposing slots **606** and into the pair of opposing channels **408** to allow for moving the cartridge **102** between the stored and dispensing positioned and, ultimately, sliding the vial **602** within the shell **600** for dispensation of the medication.

[0113] The pair of opposing slots **606** are formed through the surface of the shell **600** to allow access to the channels **408** therein. These slots **606** are designed with precise measurements to fit and accommodate corresponding elements, namely, a distinct pair of opposing protrusions **406**. The opposing slots **606** are carved or formed into the shell **600** and aligned in such a way that they maintain a clear, unobstructed path for the protrusions **406** to pass through them with ease. The unique arrangement allows for the insertion of the pair of opposing protrusions **406**, which are another critical characteristic of this one embodiment. These protrusions **406**, with their matching dimensions to the slots **606**, are designed to glide smoothly through the opposing slots **606** and into the aligned opposing channels **408**. The path provided by the slots **606** guides and directs the protrusions **406** into the appropriate channels **408**. Consequently, this ensures a specific interaction between the three parts: the opposing protrusions **406** smoothly pass through the opposing slots **606**, and they are then guided into the opposing channels **408**, maintaining a suitable and secure connection.

[0114] Further, the medication cartridge **102** presents an innovative design with respect to the pair of opposing slots **606**. Each of the pair of opposing slots **606** is formed of a first slot **708**, a second slot **710**, and a third slot **712**. These slots are strategically positioned and oriented for the optimal functioning of the device. The first slot **708** in each pair begins at the bottom end **706** of the shell **600** and extends upward toward the top end **704**. It is arranged in such a way that it mandates a route which ascends from the base to a designated position along the length of the cartridge **102**, otherwise referred to as the first junction **714**.

[0115] Subsequent to the first junction **714**, the structure of the shell **600** features a second slot **710** for each pair. Instead of following the upward trajectory of its predecessor, this second slot **710** takes on a unique characteristic of its own by providing an angled path that descends from the first junction **714**. This downward movement proceeds until it reaches a designated point along the body of the shell **600**, referred to herein as the second junction **716**. Moreover, the

slot design of the shell **600** reverts back to an upward direction in the subsequent and final arrangement of each pair—the third slot **712**. Commencing from the second junction **716**, this slot ascends once again (e.g., vertically) towards the top end **704** of the cartridge. The distinct positions and paths of the slots **708**, **710**, and **712** provide a sophisticated layout that enhances the fundamental operations of this medication cartridge **102**. Notably, while the channel **408** terminates at the second junction **716** or channel terminal **718**, the third slot **712** proceeds upwards from the second junction **716**. This differentiation allows for the selective dispensing of the medication therein. This innovative structure enhances the controlled dispensing of medication, making this cartridge **102** particularly advantageous in the administration of drugs. The angles and special slot layout improves the overall efficiency and operational effectiveness of the delivery system, which in turn can broaden its potential applications in the medical and pharmaceutical fields. The precise orientation of the slots from bottom to top and their alternating directions contribute to creating a balanced and smooth transition of the medication within the cartridge **102**. The level of control provided by this inventive structure is a significant evolution in medication delivery system designs.

[0116] For further understanding, FIGS. **8A** and **8B** depict a medication cartridge **102** as being installed or otherwise inserted into the cartridge carrier **200**. As shown, the alignment features **406** (i.e., opposing protrusions) provided a keyed connection and alignment by passing through the channels **408** and slots **606** to allow the cartridge **102** to be contained within the carrier **200**. As shown in FIG. **9**, the alignment feature **406** has travelled through the first slot **708** and channel **408** to rest at the first junction **714**. In this position, the medication cartridge **102** is held within the dispensing device in a stored position. Since the alignment feature **406** is at a fixed location within the cartridge carrier **200**, and because the second slot **710** and channel **408** descend downwards at an angle from the first junction **714**, rotation of the cartridge carrier **200** necessarily forces the medication cartridge **102** upwards within the cartridge carrier **200**. These aspects are further depicted in FIGS. **10A** through **10C**.

[0117] FIG. **10A** depicts the medication cartridge in a stored position. As shown, the alignment feature **416** rests within the first junction **714**. As shown in FIG. **10B**, rotation **1000** (via the gear wheel **404**) of the cartridge carrier **200** forces the alignment feature **416** to traverse through the second slot **710** and channel **408** away from the first junction **714**. In doing so and as shown, the medication cartridge **102** is lifted upwards within the cartridge carrier **200**. Further rotation **1000** of the cartridge carrier **200**, as shown in FIG. **10C**, results in the alignment feature **416** to traverse until it rests within the second junction **716** and the channel **408** terminates at the channel terminal **718**. Notably, when the alignment feature **416** is at the second junction **716**, the medication cartridge **102** is lifted from the dispensing device **100** into the dispensing position. Thus, the journey of the protrusions commences from the first junction **714**, progressing towards the second junction **716**. As a consequence of this carefully guided pathway, the medication cartridge **102** experiences an upward shift. This movement is instrumental in the transition of the medication cartridge **102** from what is denoted as the ‘stored position’ to its final location, the ‘dispensing position’.

[0118] While the cartridge 102 is in the dispensing position, the vial 602 therein is in the expanded state and ready to be compressed to dispense medication. As shown in FIGS. 11A through 11C, although the channel 408 in the vial 602 has terminated, the slot in the shell 600 continues upwards as the third slot 712. Thus, when the medication cartridge 102 is in the dispensing position, a user can selectively press down on the medication cartridge 102 to cause the vial 602 to slide within the shell 600. As shown in FIG. 11C, due to the stepped inner wall 1100 and end of the channel 408, the vial 602 is held in place by the alignment features 406 and only the outer shell 600 can travel downward when pressed down, causing the shell 600 to slide downward over the vial 602 and allowing the pump assembly to be actuated and spray/dispense the medication. This feature is further illustrated in FIGS. 12A and 12B. As shown, pressing downward 1200 on the medication cartridge 102 causes the alignment features 406 to travel upwards through the third slot 712, thereby forcing the shell 600 downward while holding the vial 602 in place. While the shell 600 is forced downward, it compresses the vial 602 from the expanded to compressed state. The transition from the expanded to compressed state activates the pump assembly to force medication 1202 from the vial 602 and through the dispensing aperture 604, similar to a squirt of medication. The pump assembly can be configured as a spring-loaded pump mechanism that lifts the shell 600 after being compressed, thereby returning the alignment features 406 to the second junction 716. Once dispensed, the dispensing device 100 and associated activator mechanism 300 can be activated to turn the gear wheel 404 in an opposite direction as previously rotated. Such a rotation lowers the medication cartridge 102 back to the stored position as the alignment features 406 are forced back to the first junction 714 (shown in FIG. 10A).

[0119] In one aspect, the device 100 is formed to sense once a predetermined number of squirts (e.g., one, etc., as prescribed) have been initiated so that it automatically activates the activator mechanism 300 to return the medication cartridge 102 back to the stored position, thereby retracting the medication cartridge 102 and inaccessibility after user. For example, magnetic sensors, a trigger, electrical sensors, light sensors, or any other means for determining if medication has been dispensed from the vial 602 can be included within the device 100. As a non-limiting example, a magnet can be positioned at the bottom of the shell 600 while a magnetic reed switch is similarly positioned at the bottom of the cartridge carrier 200. Pressing the medication cartridge 102 and its shell 600 downward within the cartridge carrier 200 brings the magnet into close proximity of the magnetic reed switch to notify the associated electronics (integrated circuit, etc.) that a single squirt has been dispensed. After the predetermined number of squirts, the activator mechanism is activated to rotate the cartridge carrier 200 and return the medication cartridge 102 to the stored position to prevent further and unauthorized use. In another aspect, the device 100 can include any necessary components (integrated circuit, etc.) to allow for a timed configuration in which the medication cartridge 102 is held in the dispensing position. For example, the device 100 can be programmed such that after authentication and movement of the medication cartridge 102 into the dispensing position, it is returned to the stored position after a predetermined or preprogrammed amount of time (e.g. 2 seconds, etc.).

[0120] As noted above, a unique design feature allows for the vial 602 to exist in two distinct states; namely an expanded state and a compressed state. The transition between these two stages is critical to the functionality of the device. When it moves from the expanded state towards the compressed state, the incorporated pump assembly 700 is triggered into action. This incorporation of a pump assembly 700 and its associated operation indicates an upbeat sophistication of the one embodiment, going beyond traditional static medical dispensing devices. The purpose of this actuation is specifically to ensure the controlled release of the medicine within the vial 602. On the actuation of the pump assembly 700, the medicine 1202 is pushed out of the vial's 602 outlet. This outflow proceeds under the controlling and moderating role of the pump assembly 700, guaranteeing a controlled and managed release of the medicine.

[0121] In summary, the present disclosure provides an innovative medication dispensing device 100. The core of this medication dispensing device 100 revolves around a unique cartridge carrier 200 design, integrated with a pair of opposing protrusions 406 and a uniquely designed medication cartridge that is functionally designed to operate based on the positioning of the protrusions 406 after user authentication. As noted above, in one aspect, the device 100 includes an authentication component 106. Upon successful authentication, the authentication component 106 triggers the activator mechanism 300. In this context, the activation entails the initiation of moving the medication cartridge 102. When activated, it can efficaciously induce the cartridge carrier 200 mechanism to actuate, progressing the inserted medication cartridge 102 from a stored position to a dispensing position. Maximal effectiveness is ensured as the activator mechanism 106 is engendered only after a successful user authentication process, thereby verifying that the medication is dispensed to the appropriate user. This sequence not only ensures secure and controlled dispensing of the medication but also minimizes errors and the potential for unlawful access. In one aspect, the interconnectedness between the authentication component 106 and the activator mechanism 300 forms the crux of this system, thereby augmenting the overall safety and operation of the dispensing device 100. Moreover, this intricate mechanism, in coordination with the user and the cartridge carrier 200, ensures that the process of moving the medication cartridge 102 from a stored position to a dispensing position is not only smooth but also efficient and secure.

[0122] Thus, in one aspect, a purpose of this inventive operation is to modify the positioning of the medication cartridge 102 from a state of storage to a dispensing state, thereby facilitating the delivery of medication in a controlled, efficient, and timely manner. The operational process involved in use of the dispensing device 100 advances through several stages, starting with the activation of the activator mechanism 300. The initiation of this mechanism 300 leads to a rotational motion in the cartridge carrier 200, a critical movement that fosters the maneuvering of the pair of opposing protrusions 406 (i.e., alignment features). This rotation is precisely designed to stimulate this movement, making the entire process efficient and seamless. The pair of opposing protrusions 406 are guided in their motion by a pair of opposing slots 606 in conjunction with a pair of opposing channels 408. These channels 408 and slots 606 function in harmony, not only providing a safe passage but also controlling the direction of the movement of these

protrusions 406 while guiding the medication cartridge 102 between a stored position and a raised dispensing position. The unique features of the slots 606 and channels 408 also allow a user to compress the vial 602 within the medication cartridge 102 to dispense medication 1202.

[0123] As noted above, the present disclosure provides two example embodiments of the medication cartridge, both of which can be used with the handheld medication dispensing device. While FIGS. 1A through 12B were directed to a first embodiment of the medication cartridge 102, FIGS. 1B and 13A through 16E depict and are directed to a second embodiment of the medication cartridge 101. It should be noted that the dispensing device 100 as described herein can be also configured and/or programmed to work with the second medication cartridge 101 according to the necessary features/operations as described herein. Further, in some aspects, each of the features of the first medication cartridge 100 are equally applicable to the second medication cartridge 101 with the exception of the features specifically described below.

[0124] Importantly and as shown in FIGS. 13A and 13B, in addition to all of the components and features as described above, the dispensing device 100 includes the cartridge carrier 200 with a keyed connector (i.e., alignment feature or pair of protrusions 406) that is designed to align with and operate the medication cartridge 101. In addition to ensuring the correct placement and alignment of the medication cartridge 101 within the dispensing device 100, the protrusions 406 are operable for facilitating dispensing/pumping of the medicine within the medication cartridge 101.

[0125] FIGS. 14A through 14C depict the medication cartridge 101 of the second embodiment. As shown, the medication cartridge 101 includes a shell 1400 that conceals a medicine vial 1402 slidably positioned therein. The proximal or top end 1410 of the shell 1400 includes a pump cap 1404 with a dispensing aperture 1406. Importantly, the top or proximal end of the vial 1402 includes a pump mechanism 1408 integrated therein that is operable for pumping fluid from the medicine vial 1402 and out of a dispensing aperture 1406. The pump mechanism 1408 is any suitable mechanism or device that is operable for selectively pumping fluid from the vial 1402 and out of the dispensing aperture 1406. As a non-limiting example, the pump mechanism 1408 is similar to as described and illustrated in FIG. 7H. However, instead of pressing down on the medication cartridge 100 as shown in FIGS. 12A and 12B, in the second embodiment, the medicine vial 1402 is forced upward within the shell 1400 to compress the pump mechanism 1408 and draw/force medicine or fluid from the vial 1402 and out of the aperture 1406.

[0126] Compressing the pump mechanism 1408 within the medication cartridge 101 is accomplished with the unique slotting and channeling on the shell 1400 and vial 1402, respectively. In this aspect, the bottom end 1412 of the shell 1400 has one or more rising slots 1414 (e.g., a pair of opposing slots) that pass through the shell 1400 and rise from the bottom end 1412 toward the top end 1410. Similarly, the bottom end 1416 of the vial 1402 includes or more rising channels 1418 (e.g., pair of opposing channels) that are formed within an exterior surface of the vial 1402 and that align with the one or more slots 1414. Each of the slots 1414 and channels 1418 rise from the bottom ends 1412 and 1416 until they reach a common apex 1422, at which point they directionally diverge. At the apex 1422, the rising slots

1414 turn laterally and continue passing through the shell 1400 as horizontal or lateral slots 1424. Alternatively, at the apex 1422, the rising channels 1418 alter course and descend at an angle (below the lateral slots 1424) as downward descending channels 1426. It should be noted that although the illustrations only depict slots and channels on one side, the invention is not intended to be limited thereto as similar opposing slots and channels can be formed on the other side of the shell 1400 and vial 1402 that engage with the opposing protrusions 406. Further, each of the downward descending channels 1426 terminate in a channel terminus 1430 which coincides with the end of the pumping motion provided by the pump mechanism 1408.

[0127] Thus and as was the case above, the rising slots 1414 and lateral slots 1424 are formed through the surface of the shell 1400 to allow access to the channels 1418 and 1426 therein. The slots 1414 and 1424 are formed through the shell 1400 and aligned in such a way that they maintain a clear, unobstructed path for the pair of protrusions 406 to pass through them with ease and into the aligned opposing channels 1418 and 1426. For example and as shown in FIGS. 15A and 15B, the medication cartridge 101 is formed to slide into the dispensing device 100 by aligning the protrusions within the rising slots 1414 and associated rising channels 1418.

[0128] Once inserted within the dispensing device 100, the medication cartridge 101 can be selectively activated to pump medicine from the vial 1402. The dispensing device 100 can be activated using any suitable technique and/or process as described above to cause the motor to activate and, in doing so, rotate the gear wheel and cartridge carrier.

[0129] However and to be contrasted with the design referenced above, in this aspect and as shown in FIGS. 15A and 15B, the dispensing device 100 can be modified slightly to include a rotation lock 1500. The rotation lock 1500 is any mechanism, device, or configuration that allows for rotation of the cartridge carrier 200 while preventing rotation of an inserted medication cartridge. As a non-limiting example, the rotation lock 1500 can be a protrusion or other form or marking that is attached within the dispensing device 100 and protrudes upward into the cartridge carrier 200 to lock against a portion of the medication cartridge 101 (e.g., against the shell and/or vial). Alternatively and as yet another example, the rotation lock 1500 can be a depression formed within the dispensing device and below the cartridge carrier 200 that engages with a protrusion that projects from the bottom of the medication cartridge 101. In one aspect and as shown in FIG. 13B, a hole 1300 or other aperture can be formed in the bottom of the cartridge carrier 200 (and associated gear wheel) to allow the rotation lock 1500 to pass therethrough and lock against a portion of the medication cartridge 101. In this aspect and as shown in FIG. 14A, a keyed marking 1440, such as an indentation or other shaping, can be formed in the bottom of the medication cartridge 101 (e.g., on the bottom exterior or bottom side exterior of the shell 1400) to allow the rotation lock 1500 (of FIGS. 15A and 15B) to lock against the keyed marking 1440 and prevent rotation therebetween.

[0130] Further, a slidable coupling is included to prevent rotation between the shell 1400 and vial 1402. The slidable coupling is any suitable mechanism, device, or configuration in which the vial 1402 is prevented from rotating with respect to the shell 1400, yet allowed to slide within the shell 1400. As a non-limiting example, protruding inward from

the interior walls of the shell 1400 are a pair of elongated ridges 1442 while the exterior wall of the vial 1402 includes an elongated channel 1444. When the vial 1402 is within the shell 1400, the ridges 1442 align and slide within the elongated channel 1444, yet prevent rotation between the vial 1402 and shell 1400. Thus, in this aspect, locking of the lock 1500 against the keyed marking 1440 prevents rotation of the medication cartridge 101 itself when the cartridge carrier 200 is actuated and caused to rotate, while still allowing the vial 1402 to slide vertically within the shell 1400.

[0131] As shown between FIGS. 16A and 16E, rotation of the cartridge carrier 200 in a first direction causes the vial 1402 to rise within the shell 1400 from a stored position (FIG. 16A) to a full pump position (FIG. 16C) and activate the pump mechanism 1408, thereby dispensing medicine from the medication cartridge. This process is enabled by the unique slots, channels, and keyed marking.

[0132] As shown in FIG. 16A, when the medication cartridge 101 is positioned down and into the dispensing device and associated cartridge carrier 200, the protrusions 406 pass through the rising slots 1414 and rising channels 1418 until the protrusions 406 come to rest at the apex 1422. In use, activation of the dispensing device causes motor to activate and rotate the cartridge carrier 200 in the first direction. As shown in FIG. 16B, as the cartridge carrier 200 rotates, the protrusions 406 turn laterally and pass through the lateral slots 1424. Notably, because the downward descending channels 1426 are angled downward and away from the apex 1422, continued rotation of the protrusions 406 within the descending channels 1426 causes the vial 1402 to rise upwards 1600. Forcing the vial 1402 upwards compresses the pump mechanism 1408 to activate the pump mechanism 1408 and force fluid or medicine from the vial 1402. As shown in FIG. 16C, the cartridge carrier 200 continues to rotate and, in doing so, the protrusions 406 continue to slide within the descending channels 1426 and force the vial 1402 up to compress and activate the pump mechanism 1408. This process continues until the protrusions 406 reach the channel terminus 1430, which coincides with the end of the pumping motion provided by the pump mechanism 1408.

[0133] In one aspect, upon completion of the pump cycle, the dispensing device reverses motion of the motor which causes the cartridge carrier 200 to rotate in the reverse direction (i.e., a second direction). As shown in FIGS. 16D and 16E, as the cartridge carrier 200 reverses rotation, the protrusions 406 slide within the descending channels 1426 to force the vial 1402 downwards 1602 until the protrusions reach the apex 1422. Upon reaching the apex 1422, the vial 1402 has returned to the stored position and is ready to be activated again in another pump/dispense cycle.

[0134] Finally, while this invention has been described in terms of several embodiments, one of ordinary skill in the art will readily recognize that the invention may have other applications in other environments. It should be noted that many embodiments and implementations are possible. Further, the following claims are in no way intended to limit the scope of the present invention to the specific embodiments described above. In addition, any recitation of “means for” is intended to evoke a means-plus-function reading of an element and a claim, whereas, any elements that do not specifically use the recitation “means for”, are not intended to be read as means-plus-function elements, even if the

claim otherwise includes the word “means”. Further, while particular method steps have been recited in a particular order, the method steps may occur in any desired order and fall within the scope of the present invention.

What is claimed is:

1. A medication cartridge, comprising:
 - a shell having a dispensing aperture; and
 - a vial positioned in the shell, the vial being movable between a stored position and a full pump position to dispense medicine from the vial and dispensing aperture.
2. The medication cartridge as set forth in claim 1, wherein the shell includes one or more rising slots passing through the shell and rising from a bottom end of the shell toward an apex.
3. The medication cartridge as set forth in claim 2, wherein the vial includes one or more rising channels that align with the one or more rising slots and rise from a bottom end of the vial toward the apex.
4. The medication cartridge as set forth in claim 3, wherein the shell includes one or more lateral slots that traverse laterally away from the apex.
5. The medication cartridge as set forth in claim 4, wherein the vial includes downward descending channels that descend downward at an angle from the apex.
6. The medication cartridge as set forth in claim 5, wherein the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.
7. The medication cartridge as set forth in claim 6, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.
8. The medication cartridge as set forth in claim 7, wherein the vial includes a pump mechanism with an outlet aligned with the dispensing aperture, the vial being slidable within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the vial rises within the shell to actuate the pump mechanism to pump medicine from the vial and out of the dispensing aperture.
9. The medication cartridge as set forth in claim 8, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.
10. The medication cartridge as set forth in claim 1, wherein the shell and vial include a slidable coupling to rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.
11. The medication cartridge as set forth in claim 10, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.
12. The medication cartridge as set forth in claim 1, wherein the vial includes a pump mechanism with an outlet

aligned with the dispensing aperture, the vial being i-s slidable within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.

13. The medication cartridge as set forth in claim **1**, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.

14. The medication cartridge as set forth in claim **1**, wherein the vial is slidably positioned within the shell between the stored and full pump position, the vial having a pump mechanism with an outlet aligned with the dispensing aperture and being operable for rising within the shell to activate the pump mechanism and dispense medicine from the vial and dispensing aperture when moved to the full pump position.

15. The medication cartridge as set forth in claim **14**, wherein the shell and vial include a slidable coupling to

rotationally lock the vial within the shell, thereby preventing rotation of the vial with respect to the shell.

16. The medication cartridge as set forth in claim **15**, wherein the slidable coupling includes one or more protruding ridges that protrude inward into the shell and an elongated channel formed on an exterior wall of the vial, such that the one or more protruding ridges slidably lock within the elongated channel, thereby allowing the vial to slide with respect to the shell while preventing rotation between the vial and shell.

17. The medication cartridge as set forth in claim **16**, wherein the vial is slidably within the shell between the stored position and the full pump position, such that when the vial is slid from the stored position to the full pump position, the pump mechanism is actuated to pump medicine from the vial and out of the dispensing aperture.

18. The medication cartridge as set forth in claim **17**, wherein the shell includes a keyed marking that is formed to lock against a corresponding lock formed in a dispensing device to prevent rotation of the shell with respect to the dispensing device.

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